

Analysts' response to 'AI washing' in earnings conference calls

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Working Paper

26 February 2026

Abstract

Heightened investor attention on artificial intelligence (AI) raises the possibility that firms may opportunistically exaggerate their AI activities to influence market perceptions. We study earnings conference call disclosures that reference AI and utilise the launch of ChatGPT as an exogenous AI shock. We identify a subset of firms, termed 'Speculative firms', that issue speculative AI disclosures despite having made no observable AI-related investments. While, on average, analysts and investors do not differentiate between Speculative firms and those with concrete AI disclosure – evidenced by comparable revisions to target prices and stock price performance – we show that analyst experience plays a critical moderating role. More experienced analysts systematically assign lower target prices to Speculative firms, consistent with a more discerning interpretation of technologically ambiguous narratives. We do not find that speculative firms benefit from (i) greater inclusion in AI ETFs, (ii) increased analyst coverage, (iii) higher trading volume, or experience (i) higher analyst processing costs or (ii) higher capex intensity. Collectively, these results provide novel evidence on heterogeneity in analyst responses to AI-related disclosures and shed light on the capital market consequences of AI washing practices.

Keywords: artificial intelligence, earnings conference calls, analyst forecasts, window dressing

JEL Classification: D8, D82, G14, G170, M41, O3

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1. Introduction

1.1 Market setting

The launch of ChatGPT in November 2022 sparked a sharp market reaction (Eisfeldt et al., 2024), underscoring its potential to enhance firm value and fuelling strong positive sentiment toward artificial intelligence¹ (AI) in general². In parallel, the frequency with which public companies in the US mentioned AI in their earnings conference calls (ECCs) grew substantially (Factset, 2025).

Some firms were genuinely affected by the new technology, making it relevant for management to disclose their AI activities to analysts whilst balancing the risk of knowledge spillovers (Cao et al., 2018; Glaeser and Landsman, 2021). It's also possible that some firms opportunistically invoked AI to capitalise on the broader market enthusiasm. Aligned with the literature, we refer to this (positive) exaggeration of AI activities as ‘AI washing’.

The potential risks of AI washing are so significant that the Securities and Exchange Commission (SEC) warned public companies against engaging in it in the capital markets. As former SEC Chair Gary Gensler stated: *“In essence, they should say what they’re doing and do what they’re saying”* (SEC, 2024).

1.2 Managerial and analyst incentives

Managers considering AI washing in ECCs may be motivated by the prospect of (at least) temporarily boosting their share price by persuading analysts and investors of their AI narrative (Rosano and Suh, 2024; Li, 2025). For example, they may wish to window dress poor operating performance. Thinking more defensively, managers may also be concerned

¹ According to the OECD (2019, p. 15), an AI system is defined as a “machine-based system that can, for a given set of human-defined objectives, make predictions, recommendations or decisions influencing real or virtual environments.”

² Rouf and Siddique (2023) identify three motives: lowering capital costs, improving liquidity, and signalling managerial talent.

that they will be penalised for not mentioning AI when their peers do. Managers with stock-based compensation will also benefit from bolstering their share price. Managers weigh these benefits against the risk that investors and analysts punish them, potentially harming their firm's share price. This trade-off depends on the perceived likelihood that market participants can verify a firm's true AI exposure, activities, and plans – a probability often viewed as low due to verification challenges.

Analysts play a critical monitoring role in curbing opportunistic managerial behaviour, especially during major technological shifts when uncertainty is heightened. Understanding their effectiveness in this capacity is therefore essential for both markets and regulators. Sell-side analysts build reputations on accurate forecasts and profitable recommendations (Jackson, 2005; Fang and Yasuda, 2009) prior research shows they face career penalties for making poor predictions (e.g., Mohammad and Nathan, 2008; Hong and Kubik, 2003). These incentives drive analysts to detect opportunistic AI disclosures and adjust their forecasts and recommendations accordingly.

1.3 AI exposure signals

In practice, identifying genuine AI exposure is difficult due to technological, firm, and analyst-specific factors. The intangible and often 'back office' operationalisation of AI obscures observation, while its diverse sub-technologies – with varying maturity and impact – complicate signal interpretation. As a result, analysts may mistakenly conflate effects across different AI domains. In practice, many firms simply use the terms "artificial intelligence" and "AI" in their disclosures, without being specific about the sub-technology; thus, increasing the noise in these firms' signals, making interpretation that much harder. Indeed, Christensen et al (2022) find that firm disclosures on specialised subjects increase analysts' cognitive load, potentially creating dissonant viewpoints among them.

It is well-documented that firm-level AI adoption (e.g., Hartley et al., 2025; Babina et al., 2024; McElheran et al., 2024) and, more generally, AI exposure (Eisfeldt et al. 2025) vary widely across industries, within industries, and by firm size. As a result, industry-level data offer limited guidance. Analysts must instead attempt to assess firm-specific drivers – some of which are unobservable – such as AI readiness (Uren and Edwards 2023).

Uncertainty also surrounds the sources of value firms may derive from AI. Capital markets view future gains as stemming from labour substitution (Eisfeldt et al. 2025), yet evidence of returns to date points instead to product innovation (Babina et al. 2024). While Barney and Reeves (2024) argue that AI creates no new differentiation, Liu et al. (2025) find heterogeneous effects within industries. Analysts must therefore grapple with uncertainty over the sources, scale, timing, and likelihood of firm-level AI benefits.

1.4 Efficacy of analysts' response

A central question in assessing the capital market consequences of AI disclosure is whether analysts can efficiently and uniformly interpret technologically complex narratives. The literature suggests that analysts' responses to potential AI washing may be influenced by biases, limitations in analysts' capabilities, and heterogeneity in analysts' experience.

The literature documents several biases that may inhibit analysts' ability to correctly identify opportunistic AI disclosures. For example, analysts may strategically favour a company if they want to 'curry favour' with their managers (Zhang and Wei 2020). Analysts may be distracted, for example, by concurrent earnings announcements (Driskill et al., 2020), which can diminish their information-processing capacity (Pantzalis and Ucar, 2014; DellaVigna and Pollet, 2009). These effects are amplified when high amounts of misinformation have to be processed alongside genuine information, the 'Firehose of Falsehood' (Paul and Matthews 2016). Similarly, Elkemali (2023) finds that information uncertainty leads analysts to exhibit

optimism – overreacting to positive predictions and underreacting to negative ones. This optimism is particularly pronounced for high-tech firms and increases with the forecast horizon – paralleling the development of AI.

Analysts may lack the skills to interpret disclosures when uncertainty is high (as is the case with a new, evolving, and heterogeneous technology like AI). Analysts have historically struggled to accurately predict technological developments (Vinsel 2023), aligning with Amara's Law (Dinh et al., 2015) due to uncertainty about their impact. Analysts also find it challenging to account for frictions associated with deploying new technologies, which may delay or limit the realisation of benefits versus expectations (Bresnahan and Greenstein, 1996; Brynjolfsson et al., 2019).

AI-related disclosure can, on average, improve analysts' information environment (Dao et al., 2025). However, these aggregate effects may conceal systematic differences in analysts' abilities to evaluate strategically framed and exaggerated claims. Prior evidence documents substantial heterogeneity in analyst skill and judgment, with forecast and valuation outputs reflecting analyst-specific attributes rather than mechanical updating (Mikhail et al., 1997; Bradshaw et al., 2013; Crane and Crotty, 2020).

We contend that analyst experience is a key moderating factor. Through repeated exposure to managerial forecasts and innovation cycles, experienced analysts are better positioned to assess the reliability of forward-looking technological narratives (Mikhail et al., 1997) and to process innovation-related information more effectively (Tan et al., 2019; (Mohammadi et al., 2015)). Alternatively, experience may also foster greater scepticism toward disclosures lacking operational substance – an effect likely amplified during periods of rapid technological change (Benner and Ranganathan 2008).

1.5 This paper

Analysts play a key role in monitoring corporate disclosure, a task that becomes even more crucial under conditions of heightened uncertainty. Yet their effectiveness in such contexts is little understood. This paper fills that gap by providing systematic evidence on analysts' responses to ambiguous signals following a technological shock.

The setting we use is the financial reporting window before and after the launch of ChatGPT on 30 November 2022. Critically, this event elevated AI to being one of the most critical business issues discussed by firms, investors, and analysts, and therefore, its importance as a potential discussion point on ECCs³. Hence, our main events of interest are firms' first ECCs after the launch of ChatGPT (the "initial ECC"). We apply textual analysis techniques to the transcripts of these calls to identify the ECCs in which management mentioned AI. We find that AI was mentioned in 17% of the available ECCs, resulting in a sample of 549 ECCs (firms).

We seek to examine analysts' responses to the heterogeneity of firms' AI disclosures in this period of heightened uncertainty and potential misinformation. A critical empirical challenge is distinguishing between opportunistic firms and those with substantive AI activities. To do this, we utilise ChatGPT to identify AI disclosures that are 'speculative' or not (following Cheng et al., 2019). Speculative disclosures are insubstantive and typically provide vague future plans for implementation. We combine this firm-level disclosure attribute with the firm's AI investment (following Babina et al., 2024) to classify a firm as either 'Speculative', where the firm uses speculative language and has not undertaken any AI investment, or 'Existing'⁴ (or a 'genuine' firm) otherwise.

³ This discontinuity is illustrated in Appendix 1, which shows a rapid increase in firm disclosures, including the term "ChatGPT."

⁴ Following the naming convention used by Cheng et al. (2019)

We validate our measure by showing that Speculative (Existing) firms had a lower (higher) prevalence of several AI-related firm characteristics (e.g., inclusion in exchange-traded funds (ETFs)).

We find that, on average, analysts and investors do not distinguish between speculative firms and those with concrete AI disclosure. Speculative disclosure firms benefit from higher target prices and stock price performance similar to that of firms with concrete AI disclosure.

Cross-sectional tests reveal that, conditional on analyst experience, this aggregate picture changes markedly. Per our contention, we find that more experienced analysts systematically assign lower target prices to firms engaging in speculative AI disclosure, consistent with stronger discounting of technologically ambiguous narratives. Given that target prices represent analysts' explicit mapping of expected cash flows into firm value (Bradshaw et al., 2013), these findings suggest that experience disciplines the valuation impact of speculative AI disclosure. The cross-sectional tests also find that analyst biases are not strong moderators of this aggregate finding. Collectively, the evidence highlights that capital market responses to emerging technology disclosure are shaped not only by firm behaviour but also by heterogeneity analyst's responses.

We do not find that speculative firms benefit from (i) greater inclusion in AI ETFs, (ii) increased analyst coverage, (iii) higher trading volume, or experience (i) higher analyst processing costs or (ii) higher capex intensity.

These are important insights for firms, regulators (especially the SEC), investors and analysts to consider. Our findings contribute to the growing evidence that some firms engage in AI washing behaviour. Our paper complements the existing literature by studying analysts' reactions to suspected AI washing.

The rest of the paper proceeds as follows. Section 2 outlines the connection between this paper and the related literature. Section 3 describes the data and methodology we have used. Section 4 presents the empirical results and robustness tests. Section 5 concludes.

2. Related literature and hypothesis development

2.1 *[x] Washing*

The conjectured phenomenon of “AI washing” follows in a long line of theme or shock-based attempts at theme-based, opportunistic image management. These include: ‘purpose washing’ (Findlay and Moran, 2019), ‘diversity washing’ (Baker et al., 2024), ‘blockchain washing’ (Cheng et al., 2019; Wang et al., 2024), ‘CSR washing’ (Attig et al., 2023), climate change risk exaggeration (Bratten and Cheng, 2024) and going further back, ‘dotcom washing’ (Cooper, Dimitrov, and Raghavendra Rau, 2001).

2.2 *AI washing*

There is a small yet emerging literature on the subject of AI washing. This literature generates insights into three related areas: the prevalence of AI washing, the stock price-based incentives for the practice, and the firm determinants of AI washing.

There is moderate to strong evidence in the literature of the prevalence of AI washing. Rosano and Suh (2024), (Barrios et al., 2025), Li (2025) and Yao et al. (2025) find evidence for the practice, whereas Cao et al. (2024) find a strong correlation between AI disclosure and a firm’s actual AI activities. Donelson et al. (2025) and Zeyang Ju (2025) also find evidence of a phenomenon they call “AI masking” (“AI hushing”). This is the tendency for some firms to understate their AI activities. Donelson et al. (2025) find that firms undertake AI masking to avoid scrutiny of the social and environmental impacts of AI investments. Whereas Zeyang Ju (2025) reports that AI hushing is a strategy to avoid proprietary costs. The strength of this

evidence is somewhat surprising, given that the majority of these papers analyse mandatory disclosure (10-K and 8-K), which is subject to audit and/or regulatory constraints and scrutiny.

Consistent with the above general finding, the literature provides strong evidence of firm-level incentives for AI washing, as reflected in bolstered stock returns. Both Rosano and Suh (2024) and Li (2025) find that when firms mention AI in their quarterly earnings announcements or conference calls, investors react more strongly to their earnings; thus, providing an incentive for firms to engage in AI washing.

Several papers provide insights into the firm-based determinants of AI washing. There is little consensus on the key drivers, which is consistent with the evidence on prevalence. On the one hand, Barrios et al. (2025) find that suspected AI washers tend to be smaller, less innovative and in non-AI-intensive industries. On the other hand, Zeyang Ju (2025) finds that technology peer pressure is higher amongst AI washers, which drives greater AI washing behaviour at the extremes of the innovation leadership spectrum. Other significant firm determinants identified in the literature include lower cost of equity and higher stock liquidity (Ju, 2025), as well as greater financing needs, weaker board governance, and increased investment funds (Donelson et al., 2025). Finally, Al Haddi (2024) argues that organisational cultural drivers can increase the likelihood that some CEOs engage in AI washing – for example, a lack of technical literacy, short-termism, and a ‘fear of missing out’.

2.3 Heterogeneity in analyst performance

In addition to firm-level variation, analyst heterogeneity is central to understanding responses to AI washing. Crane and Crotty (2020) demonstrate substantial heterogeneity in analyst skill, indicating that performance is shaped by analyst-level attributes as much as by firm characteristics. Specifically, Bradshaw et al. (2013) find that variation in target prices reflects

analyst-specific judgment rather than mechanical updating. That said, Beunza and Garud (2007) suggest that enhanced expertise can alter attention and ‘interpretive frames’, potentially amplifying selective processing of strategically framed technological disclosures.

In our setting, analysts must translate disclosures of complex technologies into expectations about future cash flows. Evidence indicates that expertise in innovation-related information systematically shapes how such disclosures are processed and incorporated into forecasts (Tan et al., 2019; Mohammadi et al., 2015). Technological proximity between analysts and firms further enhances performance, consistent with more efficient interpretation of technical narratives (Yu et al., 2025). At the same time, expertise may also shape attention and framing. In periods of radical technological change, analysts exhibit systematic biases, including a preference for particular firm types (Benner and Ranganathan 2008). Thus, technological knowledge may either improve analysts’ ability to detect overstated claims or reinforce selective interpretation when disclosures are strategically framed.

Evidence from adjacent “washing” contexts suggests that analyst monitoring can discipline opportunistic disclosure, but the strength of this effect varies with analyst presence and characteristics, consistent with heterogeneous scrutiny rather than uniform vigilance (Wang et al., 2024).

Evidence from AI disclosures relating to analysts remains very limited. (Dao et al., 2025) examine the relation between AI disclosures and analysts’ information environment. They show that greater AI-related disclosure increases analyst following, reduces forecast dispersion, and improves forecast accuracy, consistent with enhanced information efficiency. Their findings suggest that AI disclosures are, on average, informative to analysts. However, the study does not examine analysts’ responses or the effects of strategically framed or speculative disclosures.

Taken together, the literature implies that while AI disclosures may improve the information environment on average, analyst heterogeneity remains central when disclosures are technologically complex and strategically framed. Moreover, the existing literature does not test the success of an AI washing strategy from the firm's perspective by directly testing analyst and investor reactions to AI disclosures from suspected AI washers. This is the focus of our paper.

2.4 Hypothesis development

Analyst reaction

Consistent with the literature, we predict that Speculative firms employ AI washing to mimic Existing firms. By appearing to be Existing firms, they hope that analysts will associate them with being more AI-exposed than they actually are. In doing so, they hope to elicit a more positive response than they otherwise would (similar to that of an Existing firm). They calculate this will result in improved earnings forecasts, target valuations, and/or stock ratings. However, it is uncertain ex-ante whether such a strategy will be successful.

On the one hand, there are reasons to expect analysts to be impervious to misdirection. Being professional information intermediaries, analysts are expected to process all information rationally and efficiently and are not prone to unexpected distractions (Israeli et al., 2022). Furthermore, they would be unresponsive to firms providing vague, exaggerated, or unsubstantiated information because this would be considered uninformative (Beaver, 1968).

On the other hand, analysts are shown to exhibit irrational behaviour due to strategic incentives (Kim et al., 2023; Bosquet et al., 2015). They are also subject to biases such as overconfidence (Bosquet et al., 2015), inertia (Bernhardt et al., 2016) and confirmation bias (Cai et al., 2022). Kim et al. (2024) quantify the extent of this irrationality by comparing analyst forecast accuracy to that generated by AI. Indeed, previous technology-based hype cycles, such as the dotcom bubble (Cooper et al., 2001; Goldfarb and Kirsch, 2019) and the bitcoin "mania" (Cheng et al., 2019; Wang et al., 2024), demonstrate that firms have successfully misdirected market participants via corporate announcements.

ECCs are an intriguing setting that amplifies the uncertainty of the manager-analyst dynamic. As an unaudited format, firms have greater scope and flexibility to script deep-dive presentations and to be more 'creative' in their messaging. However, the spontaneous nature of the Q&A creates some jeopardy for managers, exposing them to the potential of revealing their true intentions (Larcker and Zakolyukina, 2012; Matsumoto et al., 2011). ECCs are also, arguably, less open to manipulation by firms trying to influence users of textual analysis, unlike 10-K filings (Cao et al., 2023).

If Speculative firms successfully deceive analysts through AI washing, we expect analysts' reactions after the initial ECC to be the same for both types of firms. Therefore, we formulate our central hypothesis in the null form:

H1: Analyst target price revisions following the initial ECC are the same for Speculative and Existing firms.

We posit that accumulated experience shapes analysts' assessments of the credibility of speculative technological disclosures. Through repeated exposure, experience enables analysts to evaluate the ex post reliability of managerial narratives and the track record of forward-looking claims (Mikhail et al., 1997). Such experience also enables analysts to better

discern the credibility of technologically ambitious statements (Bradley et al., 2017). In such settings, more experienced analysts may apply greater discounting to the speculative elements of projected value creation. Because target prices reflect analysts' explicit mapping of expected cash flows into firm value (Bradshaw et al., 2013), stronger discounting of speculative narratives should manifest in more negative target price revisions following the initial ECC.

H2: More experienced analysts reduce their target prices for Speculative firms following the initial ECC.

Investor reaction

We complement our test for analysts' reactions by providing a pseudo-robustness test for our central hypothesis, by also testing investor reaction. Classical finance and accounting theory suggests that the responses of investors should be the same as those of analysts, given that they have access to the same information. It also provides an opportunity to explore any differences in the reaction, which may offer some interesting insight into this behaviour. If Speculative firms successfully deceive investors through AI washing, we expect investors' reactions after the initial ECC to be the same for both types of firms.

In the context of analysts, Bushee and Huang (2024) discuss the concept whereby analysts may have second thoughts after an initial upgrade (downgrade), due to inefficient processing. This results in analysts revising their view downwards (upwards), consistent with Yao et al., (2025). We test for a mirror of this behaviour in the context of investors by testing for cumulative abnormal returns (CAR) across both an initial 3-day [-1, +1] window, but also a 7-day window [-1, +5] to test for any permanence in the reaction and to capture any potential 'regret correction'.

Second-order effects

To deepen our analysis of Speculative firms' incentives, we explore the possibility that they may experience second-order benefits and costs from their disclosure strategy. For example, firms may benefit from greater inclusion in ETFs, increased analyst coverage, or a rise in trading volume. At the same time, there may be unintended consequences, such as higher proprietary costs of disclosure processing for analysts or increased capital expenditure intensity – the cost of following through on their exaggerated claims about AI activities and plans.

3. Data and Methodology

3.1 Sample and data sources

Our sample is drawn from the publicly listed firms in the US. We downloaded all the ECC transcripts for the four months after the launch of ChatGPT (December 2022 to March 2023) from LSEG Workspace. This comprises 3,067 transcripts. Before analysis, 36 transcripts were mechanically flagged and excluded from the sample, leaving 3,031. The reasons for exclusion were: (i) idiosyncratic issues with formatting, which the automatic parser could not resolve, (ii) some transcripts only included a Presentation section, and (iii) where firms had held a second ECC in that timeframe due to their reporting cadence, only the initial ECC was retained. These transcripts were mechanically parsed to identify those in which firm management mentioned AI. This identified 563 ECCs for deeper analysis. Of these, data were missing for 210 firms in constructing the control variables used in the modelling. This left 353 firm transcripts, which passed basic data availability checks. This breakdown of the sample of transcripts is summarised in Table 1.

[Insert Table 1 Around Here]

We provide an analysis of the sample split between Speculative and Existing firms in Appendix 2. These results suggest that speculative AI disclosure began immediately after the launch of ChatGPT, with the rate of speculative disclosure following somewhat of a ‘random walk’ trend in the subsequent weeks as firms held their initial ECCs. This suggests that firms quickly identified the launch of ChatGPT as an opportunity to window dress, rather than gradually learning this behaviour by observing their peers over a longer period. This suggests that the decision to engage in speculative disclosure is driven more by firm characteristics and context than by the firm’s sequence in the reporting cycle. We also observe that the prevalence of speculative disclosure is lower in micro-industries classified as part of the AI value chain than in firms that are ‘end users’ of the technology.

The primary data sources for the firm-level financial, share price, and trading volume variables, as well as analyst forecast information, are Compustat, the Centre for Research in Security Prices (CRSP), and the Institutional Brokers’ Estimate System (I/B/E/S), respectively. Other sources used were: Audit Analytics for audit- and litigation-related data, the US Patent & Trademark Office for assignee-level patent data, Factset Ownership for institutional ownership data, Execucomp for data on CEO compensation, Bloomberg for corporate governance data, CRSP for data on ETF-holdings, and Babina et al. (2024) and the US Census Business Trends and Outlook Survey for firm- and industry-level AI exposure data.

3.2 *Setting*

The setting we use is the financial reporting window before and after the launch of ChatGPT on 30 November 2022. Before the introduction of ChatGPT, large language models and generative AI were largely known only to data science specialists. However, the introduction of ChatGPT, with its simple, user-prompt-based interface, rapidly expedited the broad understanding of its potential for managers and capital market participants alike. The critical feature of the setting is how it, almost overnight, elevated AI to become one of the most significant business issues discussed by firms, investors, and analysts; therefore, it is important as a potential discussion point on ECCs. The principal ECCs of interest are those that follow the first earnings conference call hosted by firms after the launch of ChatGPT. This event is critical, as we conjecture it is the primary opportunity for managers to shape analyst perceptions opportunistically. After this, analysts can educate themselves more deeply about the technology and its applications. This would allow them to learn about the nuances of which firms are most likely to benefit from (and lose to) generative AI (and AI in general) – i.e., AI exposure.

We capitalise on two key features of ECCs to measure analysts' reactions. Firstly, the unregulated and interactive nature of ECCs enables a potentially rich, in-depth dialogue between corporate managers and analysts (Bushee and Huang, 2024). Second, the relatively short time it takes analysts to revise their target price forecasts (1.78 days on average) minimises the influence of exogenous factors outside the content of the ECC.

3.3 *Text analysis of AI disclosure*

We use textual analysis to parse each transcript and identify those initial ECCs that include management discussion of AI ("AI ECCs") either in the Presentation or Questions & Answers sections. Having filtered out transcripts with formatting issues, and in common with other

contemporary studies (e.g., Cao et al., 2024; Rosano and Suh, 2024), we apply a dictionary-based algorithm to these initial ECC transcripts to identify AI ECCs. We construct a dictionary of AI-related terms (words, bigrams and trigrams) in two steps. Firstly, we combine and de-duplicate the dictionaries from two related studies (Cao et al., 2024; Jia et al., 2025). These papers were selected because their scopes span the pre- and post-ChatGPT eras, ensuring we capture terms related to both generative and non-generative AI technologies. In the second step, we supplement the list of terms generated in step one, by asking ChatGPT to select terms from definitions of AI component technologies (Giczky et al., 2022) and use cases (OECD, 2022). In the majority of cases, these terms describe use cases (e.g., edge detection, gesture detection) which are specific to AI – and therefore a term which management may potentially mention in an earnings call. We manually check these supplementary terms to eliminate generic terms (e.g. planning)⁵.

The final AI dictionary combines the words from steps one and two, deduplicates, and adds plurals of singular nouns. Appendix A provides the full ‘AI dictionary’ of words, terms, and abbreviations. AI-related terms include ‘AI’, ‘artificial intelligence’, ‘machine learning’, and ‘robotics’. We remove terms with alternative, non-AI meanings to mitigate the possibility of inadvertently capturing non-AI discussion.

After merging all data sets, we further refine our sample of firms by excluding firms without the requisite data from Compustat and CRSP. Having applied these exclusions, we arrive at a final sample of 563 AI-disclosing firms.

⁵ Un-tabulated analysis shows that words from step two only add a handful of firms to the list of those which disclose about AI. That is, firms that mention these use cases already use a term captured in the list generated by step one. Therefore, the list generated in step two better reflects the richness of the AI disclosure.

The literature provides evidence of the informativeness of AI disclosure to both investors (Rosano and Suh, 2024) and analysts (Dao et al., 2025). We check that we obtain the same result for analysts in our sample by regressing a dummy for AI disclosure on changes in target price estimates across our whole sample of disclosing firms (those disclosing about AI and those that don't). Although AI-related sentences accounted for only 2.3% of all sentences in management's disclosures in initial ECCs, we find that AI disclosures are 'informative' for analysts (see Appendix 3).

3.4 Identifying Speculative firms

To identify Speculative firms, we follow the literature (Li, 2025; Donelson et al., 2025; Barrios et al., 2025) by analysing the dissonance between firms' AI 'talk' and 'walk'. Whereas this literature typically measures AI talk by the word count of management's AI discussion, we follow Cheng et al. (2019) by examining the content of the AI disclosure itself. To do so, we ask ChatGPT to indicate which blocks of management disclosure it considers 'speculative', having provided it with AI-adapted definitions⁶ of speculative disclosure and examples⁷ (both from Cheng et al., 2019)⁸. We ask ChatGPT to 'score' each sentence separately before aggregating to the firm-level. Prior to being passed to ChatGPT, the company's name is removed from management's AI disclosure to mitigate look-ahead

⁶ "We classify a company as a Speculative (Existing) firm if its initial disclosure reveals that the firm lacks (has) a significant commitment or a meaningful track record in Blockchain technology. Speculative firms typically provide vague future plans to implement Blockchain technology. Another often used strategy is to simply mention the hire of an employee or a board member with Blockchain-related experience without providing information about the firm's future plans. In contrast, Existing firms, for example, provide information about available Blockchain products or services as well as acquisitions of established Blockchain projects or firms." (Cheng et al., 2019; p. 5902).

⁷ See Appendix A in Cheng et al. (2019).

⁸ The main part of the prompt we give ChatGPT is "You are a precise classification engine. You must classify AI-related disclosure text as Speculative (1) vs Existing (0), based exclusively on the definition below. Return only strict JSON following the requested schema. Do not include code fences or extra commentary. Definition: - Speculative (label=1): reveals lack of significant commitment or meaningful track record in AI; vague future plans; mentions hiring AI talent without concrete plans; aspirational statements without evidence. - Existing (label=0): provides specific evidence of AI products/services deployed, measurable AI use-cases, acquisitions of established AI projects/firms, or a clear operational AI track record."

bias. Given that ‘speculative’ is not well-defined in the context of corporate disclosures (unlike, for example, sentiment), we leverage the stochastic nature of GPT’s responses⁹ and allow for some variation in how ChatGPT processes the definition and examples in assessing each firm’s AI disclosure. To make the output more deterministic, we ask ChatGPT to perform this assessment, independently, five times. A firm is only considered to have disclosed speculatively if ChatGPT scored the disclosure a ‘1’ in at least four out of the five independent assessments. At the end of this process, each firm is tagged as having either issued speculative disclosure (Speculative Disclosure Indicator (SDI) = 1) or not (SDI = 0). This constitutes our assessment of the firm’s AI talk.

A firm’s AI walk is the extent to which it is genuinely and actively engaged in substantive AI activities and investments. To measure this, we follow the literature (Cao et al., 2024; Donelson et al., 2025; and use Babina et al.’s (2024) measure of AI investment (AIE), a firm-level count of AI-related employees. We split firms into terciles based on their AIE level (where the highest (lowest) AI investment firms are in tercile 1(3)).

To identify Speculative firms, we compare firms’ AI ‘talk’ and ‘walk’ and look for a significant mismatch between the two. We define Speculative firms as those whose AI disclosure was deemed speculative (SDI = 1) and were in the lowest AI investment tercile (AIE tercile = 3). We define the indicator variable $SPEC_FIRM$, which equals 1 for such firms. Firms that demonstrate substantive AI engagement and/or provide non-speculative disclosures (‘Existing firms’) are assigned a value of 0. Examples of AI disclosure from initial ECCs of both Speculative and Existing firms are provided in Appendix C.

⁹ Setting the temperature hyperparameter to 0.3

3.5 *Validating the SPEC_FIRM measure*

To validate the veracity of the *SPEC_FIRM* measure, we examine the results of cross-sectional tests using distinctive firm characteristics of Speculative and Existing firms. The results of the main set of cross-sectional tests are shown in Table 2. The table compares Speculative and Existing firms with respect to their AI activities and attributes that should be negatively (positively) associated with Speculative (Existing) firms. For example, significantly more Existing firms are included in AI-related ETFs than Speculative firms. The difference between Speculative and Existing firms is of the anticipated sign and statistically significant. These results are consistent with the notion that the *SPEC_FIRM* measure can identify Speculative firms.

[Insert Table 2 Around Here]

The results of other cross-sectional tests are provided in Appendices 4, 5 and 6. They present cross-sectional differences in linguistic characteristics, as well as in additional financial and operational characteristics, and in firms' motivational contexts and motives. Consistent with the literature, we find that Existing firms' AI disclosures are significantly more voluminous than Speculative firms. We also find that the content of Speculative firms' AI disclosures exhibits distinctive characteristics compared to those of Existing firms. Speculative firms' AI disclosures are significantly more vague, less specific, and less complex than Existing firms. Appendix 5 shows the cross-sectional variation for several relevant firm characteristics. Where there are significant differences between the two types of firms, these are, on balance, in line with our ex-ante expectations (displayed in column (1)).

The process of identifying Speculative and Existing is, by necessity, mechanical. Therefore, to illustrate the classification more tangibly, we provide examples of AI disclosure from Speculative and Existing firms in Appendix D.

3.6 *Measuring analyst reaction*

There are several ways to measure analyst reaction (e.g., earnings forecasts, target prices, stock recommendations), but not all of them are suitable for this setting, especially since the benefits of AI are likely to be observable only over the medium- to long-term. We select analyst target price estimates for three main reasons.

Firstly, target prices explicitly reflect analysts' valuation models, where qualitative disclosures (like AI strategy, adoption, or competitive positioning) are most naturally incorporated into cash flow projections, discount rates, and growth assumptions. Earnings forecasts are narrower, tied to short-term accounting outcomes, and may not fully capture the long-horizon implications of AI (which are uncertain and diffuse). Secondly, target prices vary continuously, providing a sensitive measure of whether analysts react to AI disclosures. Stock recommendations (e.g., Buy/Hold/Sell), on the other hand, are categorical, sticky, and infrequently revised, so they are less informative about incremental changes in analysts' views. Finally, target prices are explicitly forward-looking, making them well-suited to capturing how analysts assess the long-term potential of AI. Earnings forecasts can be anchored to consensus and bounded by managerial guidance, so analysts may hesitate to reflect AI-related optimism or uncertainty in them. Reputational incentives and conflicts of interest influence stock recommendations (analysts tend to avoid downgrades), limiting their responsiveness to new information.

Therefore, to measure analyst reaction, we construct a variable (ΔTP) that reflects the extent to which analysts revise their opinion of firms' target prices after being exposed to firms' AI disclosure (i.e., following the initial ECC). Specifically, ΔTP is the percentage change in the firms' mean scaled target price following the initial ECC. The change is calculated for each

analyst per firm between the target price submitted immediately prior to the initial ECC and the next target price they submit (whether the same or revised).

3.7 *Control variables*

We employ several control variables commonly used in prior literature, beginning with those that control for general firm characteristics. We account for firm size and market valuation dynamics by including *MARKET_VALUE*, defined as the natural logarithm of 1 plus the firm's market value of equity plus total assets minus the book value of total common equity. We also include *R&D* (intensity), which broadly reflects a firm's general technology orientation, appetite and potential for using AI. *R&D* intensity equals the firm's *R&D* expenditure divided by total assets. The Cumulated Abnormal Return (*CAR*) is included to control for the market reaction to firms' earnings announcements. *CAR* is calculated as the sum of abnormal returns over trading days -1 to $+1$ relative to the initial ECC. Finally, we include two ECC controls that account for observed heterogeneity in the timing and nature of non-AI management discussions in the initial ECCs, capturing any spillover effects from the non-AI disclosure. To this end, we include *ECC_TONE* and *Q4*. *ECC_TONE* is the tone used by the firm's managers across the non-AI-related parts of the ECC (Loughran and McDonald, 2016). *Q4* is an indicator variable equal to 1 if the ECC falls in the final quarter of the firm's financial year-end, and 0 otherwise. The control variables included have been filtered from a longer list of potential controls and confounders based on pairwise correlations and VIF testing. All variables are winsorized at the 1% and 99% levels to account for outliers. All variable definitions are contained in Appendix E.

AI is widely regarded as a general-purpose technology, with applications across both horizontal and vertical domains across a wide range of industries. However, there is evidence (McElheran et al., 2024) that the adoption of AI applications is heterogeneous across sectors.

Therefore, we also include industry controls based on the Fama-French 12 industry classification.

4. Empirical results

4.1 *Descriptive statistics*

Table 3 summarises the descriptive statistics for the observations in the main sample. It shows that significant variations exist in analysts' target price revisions, the linguistic characteristics of AI disclosures, and firms' financial and related characteristics. It should be noted that the mean change in analysts' target price was negative (-3.3%) in the period after this sample of ECCs.

[Insert Table 3 Around Here]

4.2 *Analyst reaction*

To test the reaction of analysts to AI washing (Hypothesis 1), we initially conduct an OLS regression [model (1)] using the following specification:

$$\Delta TP_i = \alpha + \beta SPEC_FIRM_i + X_i + \varepsilon_i \quad [1]$$

where i indexes firm. ΔTP_i is defined as the mean percentage change in analysts' target price, measured as the revision from their most recent target price issued before the initial ECC to the one issued immediately after it. $SPEC_FIRM_i$ is a dummy variable where Speculative (Existing) firms are denoted by 1(0). X_i is a vector of firm, industry and ECC controls. We employ robust standard errors.

We acknowledge that not all firms disclose about their AI activities, and, importantly, that ECCs are considered voluntary disclosures. Therefore, we need to consider the possibility of self-selection bias in our sample. Non-disclosure is likely due to two main reasons. First, if a

firm has not adopted AI or has no plans to do so, it may have nothing to disclose. McElheran et al. (2024) report how AI adoption varies by both firm size and industry. Secondly, the literature includes several examples of how firm attributes and managerial incentives are associated with voluntary disclosure strategies of firms in the areas of new products (e.g., Chu et al., 2025) and innovation (e.g., Glaeser and Landsman, 2021). In combination, we conclude that firms' choice to disclose about AI is non-random and that the sample is highly likely to be subject to self-selection bias.

To mitigate any self-selection bias in the sample, we employ the Heckman two-stage procedure (Heckman, 1979) to estimate a second model (2). In the first-stage regression, we estimate a probit model with the dependent variable equal to a dummy indicating whether a firm discloses its use of AI (*AI_ECC*). We utilise the number of AI patents granted in the previous three years¹⁰ (*AI_PATENT*) and institutional ownership, the percentage of a firm's shares held by institutional investors at the end of the most recent calendar quarter prior to the most recent fiscal year-end (*IO*), as exclusion restrictions.

A firm's decision to discuss its AI activities in the initial ECC is likely to be influenced by the number of recent AI patents it has been granted, since managers of more innovative companies are more likely to disclose (Huang et al., 2021). Drawing on evidence from the literature (Hsu et al., 2016; Nurleni et al., 2018), we also conjecture that a firm with higher institutional ownership is more likely to discuss its AI activities in the initial ECC. Therefore, *AI_PATENT* and *IO* meet the relevance and exclusion criteria. The control variables in the main specification are employed as explanatory variables in the first-stage model.

¹⁰ Each firm's three-year window is constructed to incorporate a one-month 'air gap' preceding the earnings reporting cycle prior to the launch of ChatGPT. This is to ensure that news of any AI patents granted to a firm is already incorporated into analysts' target price estimates before we measure the change in target price estimates pre/post initial ECC.

We augment the first specification [1] with the inverse Mills ratio (*LAMBDA*) calculated from the first-stage regression to generate the second stage of the Heckman model.

We report the results of the OLS regression (1) and Heckman estimation (2) in Table 4. Panel A presents the results of the models used to test analyst reaction, while Panel B displays the results of the first-stage (selection) equation of the Heckman correction models. The coefficient for *SPEC_FIRM* in model (1) is not significant at the 10% level. This finding suggests that analysts' target price revisions are unrelated to the measures of speculative disclosure. That is, the pattern of target price revisions following their initial ECCs for Speculative and Existing firms is not significantly different. The results for model (2), which corrects for selection bias, are similar. In Panel B, the coefficients for *AI_PATENT* and *IO* are positive and significant at the 1% level. This confirms that they meet the relevant criterion. The results from the second-stage equation in Panel A show that the coefficient for *LAMBDA* is not significant at the 1% level, indicating that any bias present is not significant. The coefficient for *SPEC_FIRM* in model (2) is also not significant at the 10% level, mirroring the result from the OLS.

[Insert Table 4 Around Here]

Finally, we conduct a robustness test of this result to control for variations in firm characteristics within the disclosing group, which may have biased the result in model (2). To do this, we construct model (3) using the same specification as model (2) but employing a Propensity-Score-Model-generated control sample instead of the main sample. That is, we match AI disclosing firms in our main sample (the 'treated' firms) with similar firms that did not disclose AI (the 'control' group), and then re-estimate model (2). The matching process used the nearest-neighbour method, selecting the single best ('nearest') match without replacement. Appendix 7 provides a more detailed analysis of the differences within our main

(treated firms) sample (Panel A) and between treated firms and the control group (Panel B). Panel A shows significant differences in the characteristics of Speculative and Existing firms, underscoring the need to perform a robustness test. The results of model (3) are presented in the final column of Table 4. The results from the second-stage equation of model (3) in Panel A are consistent with those of the other specifications.

Beaver (1968) suggests an alternative explanation for the null result. That is, AI disclosures are uninformative, leading analysts to discount them entirely. Logically, this would result in no difference in target price revisions between Speculative and Existing firms. However, Rosano and Suh (2024) find that investors react strongly to all forms of AI disclosures, suggesting that they are informative. On this basis, we discount this explanation.

Overall, the results from Table 4 provide evidence in support of our first hypothesis (*analyst target price revisions following the initial ECC are the same for Speculative and Existing firms*). This result is consistent with the notion that Speculative firms have successfully mimicked the disclosure of Existing firms, to the extent that professional analysts were unable to distinguish between them. That is, AI washing behaviour was both detected and effective.

4.3 Heterogeneity in analyst performance

The above results represent averages and may mask significant differences across the heterogeneity of analysts and firms. To reveal any such differences, we conduct a battery of cross-sectional tests. The most significant results from these tests are presented in the two panels of Table 5, using model (2) from Table 4 as the baseline specification¹¹. To apply the moderating effects, we interact the *SPEC_FIRM* with key analyst and firm characteristics. The table is split into four panels, each reporting the results of different types of moderating

¹¹ Descriptive statistics for the variables used in Tables 5, 6 and 7 are documented in Appendix 8.

factors. Panel A presents the results of testing the moderating effect of analyst characteristics, following Bilinski et al. (2025). The characteristics included are general analyst experience (*GENEXP*), analyst firm-specific experience (*FIRMEXP*), the number of firms an analyst covers (*NUMFIRM*), and the number of industries an analyst covers (*NUMIND*).

[Insert Table 5 Around Here]

The results in panel A in Table 5 reveal a potent AI washing effect that was previously masked by the statistically non-significant aggregate results in Table 4. The significant, positive coefficient for *SPEC_FIRM*, controlling for *FIRMEXP*, suggests that speculative disclosure was highly effective at inflating target prices for analysts with minimal firm-specific experience. However, the significantly negative interaction term indicates that analysts with more firm-specific knowledge are more sceptical of that firm's speculative claims. *NUMFIRM* and *NUMIND* are included here as proxies for cognitive workload. From this perspective, the result in panel A suggests that greater analyst workload is a significant negative moderator. That is, analysts covering more firms and industries are significantly more sceptical of a firm's speculative AI disclosure. However, another way to consider *NUMFIRM* and *NUMIND* is that they represent the breadth of exposure to firm disclosure. When considered this way, the result can be interpreted as meaning that analysts who benefit from broader exposure to different firms and industries are more likely to penalise speculative disclosure. This suggests that greater breadth offers analysts additional perspective when considering the potential impact of new technologies. The result for *GENEXP* is only marginally significant, suggesting that the general experience level (the difference between junior and senior analysts) plays a lesser role.

Taken together, the results from Panel A of Table 5 provide strong support for our second hypothesis (*More experienced analysts reduce their target prices for Speculative firms*

following the initial ECC). This finding is consistent with the notion that more experienced analysts can detect speculative firms and, as a result, revise their target prices downward more sharply.

Panel B presents significant cross-sectional test results relating to analyst biases and firm characteristics: analyst optimism (*OPTIMISM*) and earnings surprise (*SUESCORE*).

Intuitively, we expect analysts who issue more optimistic forecasts than their peers to be more likely to take speculative disclosures at face value, resulting in a significantly positive coefficient on the interaction term with *OPTIMISM*. However, the result in panel C is counterintuitive (showing a significantly negative coefficient). There are several candidate explanations for this result; we present two here. Firstly, there is mean reversion. That is, optimistic analysts may have entered the ECCs with inflated expectations, but when firms provided only speculative disclosures, these analysts revised their expectations downwards, more in line with their peers. Secondly, if optimistic analysts are more deeply invested than their peers in the firm's specific fundamentals and long-term growth prospects, penalising speculative AI claims may allow them to use this correction to maintain the long-term credibility of their strategic 'buy' position. This dissonance in analyst outputs is consistent with Iselin et al.'s (2021) finding that incongruent analyst outputs often reflect rational responses to information, rather than strategic biases or low-quality outputs.

The result from the *SUESCORE* cross-sectional test provides moderately strong evidence that a firm's actual earnings performance significantly moderates analysts' response to speculative AI disclosure. The negative, though weakly significant, coefficient on the interaction term with *SPEC_FIRM* indicates that speculative disclosure appears to amplify negative reactions to earnings news or dampen positive reactions. That is, where firms have good (bad) earnings news, speculative firms experience a weaker positive (more negative)

response. These results suggest a ‘performance-based caution’ mechanism. When speculative AI disclosures are made by firms that have already delivered strong earnings surprises, analysts appear more doubtful of speculative ‘cheap talk’. They likely place greater emphasis on substantive earnings data and discount speculative future claims that lack immediate evidence. Conversely, the result implies that firms with low or negative earnings surprises may find speculative AI disclosure more effective at propping up their target prices, as analysts at the baseline level of performance are more easily deceived.

Taken as a whole, the cross-sectional results reveal the underlying heterogeneity in the average result in Table 4. Indeed, they provide significant insight into the circumstances under which AI washing had a discernible effect on analysts’ options, backed by intuitive, plausible explanations.

4.4 *Investor reaction*

As a complement to our main result, which tests analysts' reactions, we also examine investors' reactions. We calculate each firm’s CAR using a market model estimated 90 days prior to its initial ECC. We consider two event windows: $[-1, +1]$ and $[-1, +5]$, with day 0 denoting the day of the initial ECC. The index used to construct the market model is the CRSP US Large Cap Index (INDNO: 1001455) because this includes firms in the market capitalisation tier below the S&P 500 (which is more consistent with our sample). The former, 3-day window is designed to capture the initial investor reaction (in which they may have to take firms’ AI disclosures at face value). The 7-day window is intended to capture investors' more considered reactions. In addition to exploring whether investors are duped by AI washing, this approach enables us to capture any persistence in, or reconsideration of, the initial response.

In Table 6, investors show a weakly statistically significant positive reaction in the very short term (the $CAR[-1, +1]$ window). However, this is only a transient effect, as by the five-day window ($CAR[-1, +5]$), the coefficient is no longer statistically significant. This indicates that the initial investor excitement is a ‘hype’ reaction that quickly dissipates as the market processes the information further. Although the results for both analyst and investor reactions indicate evidence of AI washing, the analyst reaction is more suppressed. This is consistent with the notion that analysts, tasked with producing medium- to long-term price targets, require more concrete evidence of meaningful engagement with AI (compared with investors) before they revise their fundamental valuations. As professional, specialist information processors, analysts are better, albeit not perfect, at filtering out ‘cheap talk’ than investors.

[Insert Table 6 Around Here]

Our findings are consistent with those of Rosano and Suh (2024). They found that when firms mention AI in their quarterly earnings announcements, investors respond more strongly to the earnings, regardless of the disclosure's credibility. They attribute this behaviour to ‘regret aversion’ in investors (Qin, 2015) – the fear of missing out on investment opportunities, even when the AI disclosures are inconsistent and potentially opportunistic. Overall, this consistency with the literature provides further robust support for our central hypothesis.

4.5 *Second-order effects*

We also test whether Speculative firms experience significant second-order effects (positive or negative) following their initial ECC disclosure. We consider five effects or impacts, although it is not possible to identify whether these were intended or unintended consequences. We test whether firms benefited from being included as constituents in more AI-related ETFs (ΔAI_ETF), experienced an increase in analyst coverage ($\Delta COVERAGE$), or saw an increase in trading volume ($\Delta TRAD_VOL$) for their stock. We also test for the impacts

in terms of higher analyst proprietary costs ($\Delta DAYS$) and an increase in capital expenditure intensity resulting from having to follow through on their AI disclosure ‘commitments’ ($\Delta CAPEXINT$).

We construct separate models for each second-order effect by substituting each effect as a new dependent variable in the baseline specification (model (3) in Table 4), in place of ΔTP . Table 7 reports that none of the coefficients for the second-order effects is statistically significant at the 10% level. This suggests that firms either didn’t benefit from or weren’t affected by second-order effects stemming from their AI disclosure strategies. This leaves the main benefit to Speculative firms from their opportunistic disclosure as a bolstered share price. Overall, this finding provides evidence in support of our central hypothesis.

[Insert Table 7 Around Here]

4.6 Robustness tests

To test the robustness of our main result, we run two falsification tests: an alternative outcome test and a placebo test. The results of these tests are reported in Appendices 9 and 10 respectively. The alternative outcome test re-estimates the baseline model (model (2) in Table 4) with alternative definitions for firstly, $SPEC_FIRM$, and then ΔTP . The alternative definition of the main variable of interest, $SPEC_FIRM2$, combines alternative measures of speculative disclosure and AI exposure into a Boolean expression. The alternative measure of speculative disclosure is $SPECIFIC$, a dummy variable that equals 1 when AI disclosure mentions specific initiatives (Jia et al. 2025) and 0 otherwise. The alternative measure of a firm’s AI exposure is AI_IND , an industry-level AI exposure measure from the US Census Bureau's Business Trends and Outlook Survey (2024). Combining the two alternative measures, we derive $SPEC_FIRM2$, which equals 1 for firms with $SPECIFIC = 0$ and

AI_IND in the lower tercile. In both alternative definitions, the test yields the same (null) result, thus reinforcing the robustness of our main result.

For the placebo test, we randomise the assignment of the *SPEC_FIRM* indicator variable (‘treatment’) across the sample, constraining the random assignment to the same proportion of Speculative and Existing firms. The absence of significant effects in both the baseline specification and the placebo test suggests that the null finding in our main result is not driven by model misspecification or spurious correlations, reinforcing the robustness of our (null) main result.

Given that our result is stable to alternative specifications and falsification tests, any unobserved confounding factor would need to be stronger than the observed covariates to overturn the result, which is unlikely. Furthermore, by choosing a research design setting which utilises a quasi-exogenous shock, we leave little room for omitted variables to explain our findings.

5. Conclusion

Investors’ recent attention to the impact of AI raises the possibility that firms may opportunistically exaggerate their AI activities. We identify a subset of firms, termed ‘Speculative firms’, that issue speculative AI disclosures despite having made no observable AI-related investments. While, on average, analysts and investors do not differentiate between Speculative firms and those with concrete AI disclosure – evidenced by comparable revisions to target prices and stock price performance – analyst experience plays a critical moderating role. More experienced analysts systematically assign lower target prices to Speculative firms, consistent with a more discerning interpretation of technologically ambiguous narratives.

We do not find that speculative firms benefit from (i) greater inclusion in AI ETFs, (ii) increased analyst coverage, (iii) higher trading volume, or experience (i) higher analyst processing costs or (ii) higher capex intensity. Overall, our findings contribute to the growing evidence that some firms may be engaging in AI washing behaviour.

These are important insights for firms, regulators (especially the SEC), investors and analysts to consider. Although the setting studied in this paper is a single, relatively short event window, given that AI technologies are general-purpose and still in development, it seems reasonable to expect that similarly sized events will likely occur in the future. Practitioners' abilities to decipher the correct signal are compounded by the fact that AI is a family of technologies, which makes it easier for impacts to be conflated. Understanding signals at the level of AI sub-technologies is a potentially fruitful area for future research opportunities.

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Table 1: Sample breakdown

This table shows the breakdown of the final sample of initial ECC transcripts from the total set of ECCs available from LSEG Workspace for the period 1 December 2022 to 31 March 2023 (3,356). Before analysis, 209 transcripts were mechanically flagged and excluded from the sample, leaving 3,147. Of those, 549 transcripts were found that mentioned AI.

	Sub-totals	Transcripts removed
Total sample of ECC transcripts from Eikon (Dec '22 – Mar '23)	3,356	
Exclusions due to formatting, no Q&A segment, non-‘initial ECCs’		(209)
Initial ECC transcripts of unique firms processed for mentioning AI	3,147	
Final sample of firms mentioning AI in their ‘initial ECC’	549	
Final sample of firms mentioning AI and with the requisite data for covariates	388	

Table 2: Firm AI characteristics

This table presents the characteristics of Speculative and Existing firms in terms of their AI activities and attributes that intuitively should be associated with Existing firms – or ‘true positive’ measures of meaningful engagement with AI (and, therefore, ‘genuine’ disclosure). Column (1) provides the sign of the expected difference between Speculative and Existing firms, the result of which is shown in column (4). (2) and (3) show the mean value for each firm type. Column (4) displays the difference between these two means, and column (5) shows the p-value of this difference based on a t-test. The significance of the p-values is shown, where ***, **, and * denote the significance of the p-values at 1%, 5%, and 10% levels.

	Expected sign	Existing firms	Speculative firms	Difference	p-value
	(1)	(2)	(3)	(4)	(5)
<i>AI_ETF</i>	-	0.597	0.256	-0.341	0.015 **
<i>AI_PATENT</i>	-	0.703	0.429	-0.274	0.110
<i>AI_IND</i>	-	9.500	7.957	-1.544	0.077 *
<i>AI_FIRM</i>	-	0.003	0.000	-0.003	0.001 ***
<i>AI_VC</i>	-	0.296	0.211	-0.085	0.101

Table 3: Descriptive statistics

This table presents descriptive statistics for the variables included in the main regression results. Definitions for all variables and descriptive statistics for variables used in other tables are provided in the Appendix.

	N	Mean	Standard deviation	Minimum	Maximum
<i>Main dependent variable</i>					
<i>ΔTP</i>	388	-0.031	0.209	-0.981	1.807
<i>Variable of interest</i>					
<i>SPEC_FIRM</i>	388	0.144	0.352	0.000	1.000
<i>Control variables</i>					
<i>MARKET_VALUE</i>	388	7.988	2.134	2.327	14.606
<i>R&D</i>	388	0.084	0.109	0.000	0.685
<i>Q4</i>	388	0.624	0.485	0.000	1.000
<i>ECC_TONE</i>	388	0.483	0.293	-0.586	0.914
<i>CAR</i>	388	0.009	0.117	-0.420	0.305
<i>Exclusion restrictions</i>					
<i>AI PATENT</i>	388	0.867	1.687	0.000	9.512
<i>IO</i>	388	0.695	0.275	0.028	1.000

Table 4: Analyst reaction

This table reports the results of models which test the relationship between analyst reactions and speculative AI disclosure. Panel A sets out the models used to test analyst reaction, whereas Panel B displays the results of the first stage (selection) equation of the Heckman correction model used in models (3) and (4) in Panel A. In Panel A, the % change in analysts' target prices is the dependent variable in all models. Model (1) reports the results of estimating equation [1], where *SPEC_FIRM* is employed as the explanatory variable of interest. Model (2) is the Heckman model, which controls for self-selection bias using the Inverse Mills Ratio (*LAMBDA*) and otherwise follows the specification of model (1). Model (3) uses the same specification as Model (2) but employs a PSM-generated control sample instead of the main sample. All models include firm-, industry-, and earnings-conference call-level controls. Robust standard errors are used in all OLS models, but they are not adjusted for in the Heckman models. Panel B displays the first-stage models used to estimate unobservable values for *SPEC_FIRM* for models (2) and (3). In both panels, p-values are shown in parentheses, where ***, **, and * denote the significance of the p-values at 1%, 5%, and 10% levels.

Panel A: Outcome regressions (Second stage in Heckman)			
Dep. Var: ΔTP Sample:	Main (1)	Main (2)	Control (3)
<i>SPEC_FIRM</i>	0.037 (0.482)	0.024 (0.425)	-3.898 (0.174)
<i>MARKET VALUE</i>	-0.006 (0.324)	-0.004 (0.613)	0.450 (0.575)
<i>R&D</i>	0.127 (0.607)	-0.010 (0.942)	-10.333 (0.429)
<i>Q4</i>	-0.034 (0.117)	-0.033 (0.144)	0.675 (0.844)
<i>ECC_TONE</i>	-0.001 (0.990)	-0.072* (0.060)	-6.771 (0.107)
<i>CAR</i>	-0.033 (0.858)	0.029 (0.743)	7.189 (0.538)
<i>LAMBDA</i>		-0.005 (0.922)	31.524** (0.013)
Constant	0.087 (0.201)	0.123 (0.383)	-47.681** (0.042)

Panel B: First-stage Heckman regressions			
Dep. Var: <i>AI_ECC</i>			
Sample:	Main	Main (2)	Control (3)
<i>AI_PATENT</i>		0.189*** (0.000)	0.036 (0.244)
<i>IO</i>		-0.400*** (0.005)	0.560*** (0.000)
<i>MARKET VALUE</i>		0.108*** (0.000)	-0.002 (0.932)
<i>R&D</i>		1.152*** (0.000)	0.022 (0.943)
<i>Q4</i>		-0.060 (0.445)	0.128 (0.104)
<i>ECC_TONE</i>		0.275** (0.015)	0.023 (0.829)
<i>CAR</i>		0.371 (0.219)	0.138 (0.645)
Constant		-1.625*** (0.000)	-1.372*** (0.000)
Firm controls	Y	Y	Y
Industry controls	Y	Y	Y
ECC controls	Y	Y	Y
Model	OLS	Heckman	Heckman
N	405	1902	1965
N (selection equation)	-	388	370
Adj. R-squared	0.039	-	-

Table 5: Heterogeneity in analyst reaction

This table presents the results of tests for heterogeneity in analysts' reactions. The table is split into two panels: analyst experience (panel A) and analyst biases and firm characteristics (panel B). All tests are based on the Heckman adjusted model in Table 4 (model (2)). The % change in Target Price is the dependent variable in each case, and each model tests a specific moderating factor. Each model also includes controls at the firm-, industry-, and conference-call levels. Standard errors are unadjusted. p-values are shown in parentheses, where ***, **, and * denote the significance of the p-values at 1%, 5%, and 10% levels.

Dep. Var: ΔTP	Analyst experience			
	(1)	(2)	(3)	(4)
<i>SPEC_FIRM</i> * <i>GENEXP</i>	-0.000* (0.099)			
<i>SPEC_FIRM</i> * <i>FIRMEXP</i>		-0.001** (0.044)		
<i>SPEC_FIRM</i> * <i>NUMFIRM</i>			-0.013** (0.044)	
<i>SPEC_FIRM</i> * <i>NUMIND</i>				-0.056*** (0.002)
<i>SPEC_FIRM</i>	0.196* (0.070)	0.094** (0.042)	0.280** (0.033)	0.258*** (0.002)
<i>GENEXP</i>	0.000 (0.787)			
<i>FIRMEXP</i>		0.000 (0.587)		
<i>NUMFIRM</i>			-0.001 (0.600)	
<i>NUMIND</i>				-0.007 (0.324)
<i>LAMBDA</i>	-0.004 (0.934)	-0.003 (0.946)	-0.003 (0.948)	-0.002 (0.964)
Constant	0.111 (0.469)	0.115 (0.421)	0.119 (0.426)	0.171 (0.236)
Firm controls	Y	Y	Y	Y
Industry controls	Y	Y	Y	Y
ECC controls	Y	Y	Y	Y
Model type	Heckman	Heckman	Heckman	Heckman
N	387	387	387	387

p-values in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Dep. Var: ΔTP	(1)	(2)
Panel B: Analyst bias and firm characteristics		
<i>SPEC_FIRM</i> * <i>OPTIMISM</i>	-1.267*** (0.000)	
<i>SPEC_FIRM</i> * <i>SUESCORE</i>		-0.009* (0.079)
<i>SPEC_FIRM</i>	0.645*** (0.000)	0.036 (0.226)
<i>OPTIMISM</i>	0.283 (0.106)	
<i>SUESCORE</i>		0.001 (0.784)
<i>LAMBDA</i>	-0.003 (0.953)	-0.038 (0.470)
Constant	-0.079 (0.644)	0.221 (0.141)
Firm controls	Y	Y
Industry controls	Y	Y
ECC controls	Y	Y
Model type	Heckman	Heckman
N	354	376

p-values in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 6: Investor reaction

This table presents the results of models testing the association between investors' reactions and speculative AI disclosure. Model (1) uses the CAR for the day after the initial ECC as the dependent variable, and model (2) uses the CAR over five days from the initial ECC. The specifications of both models are the same as those used in model (4) in Table 4, apart from *SUE_SCORE* being substituted as a control instead of CAR (which is the dependent variable in this case). Standard errors are unadjusted. p-values are shown in parentheses, where ***, **, and * denote the significance of the p-values at 1%, 5%, and 10% levels.

Dep. Var:	<i>CAR</i> [-1, +1]	<i>CAR</i> [-1, +5]
	(1)	(2)
<i>SPEC_FIRM</i>	0.020 (0.208)	0.016 (0.387)
<i>LAMBDA</i>	0.002 (0.928)	0.006 (0.829)
Constant	-0.983*** (0.000)	-0.983*** (0.000)
Firm controls	Y	Y
Industry controls	Y	Y
ECC controls	Y	Y
Model	Heckman	Heckman
N	405	405

Table 7: Second-order effects

This table presents the results of models testing the association between several downstream firm outcomes and speculative AI disclosure. Each dependent variable represents the change in that measure following the initial ECC. Columns (1), (2) and (3) assess the presence of ancillary benefits (inclusion in more AI ETFs, greater analyst coverage, and higher trading volume). Columns (4) and (5) test the presence of ancillary costs or consequences (namely, an increase in analyst processing costs and higher capex intensity). Each model is a Heckman second-stage model (per model (4) in Table 4), with firm-, industry- and conference call-level control variables. Standard errors are unadjusted. p-values are shown in parentheses, where ***, **, and * denote the significance of the p-values at 1%, 5%, and 10% levels.

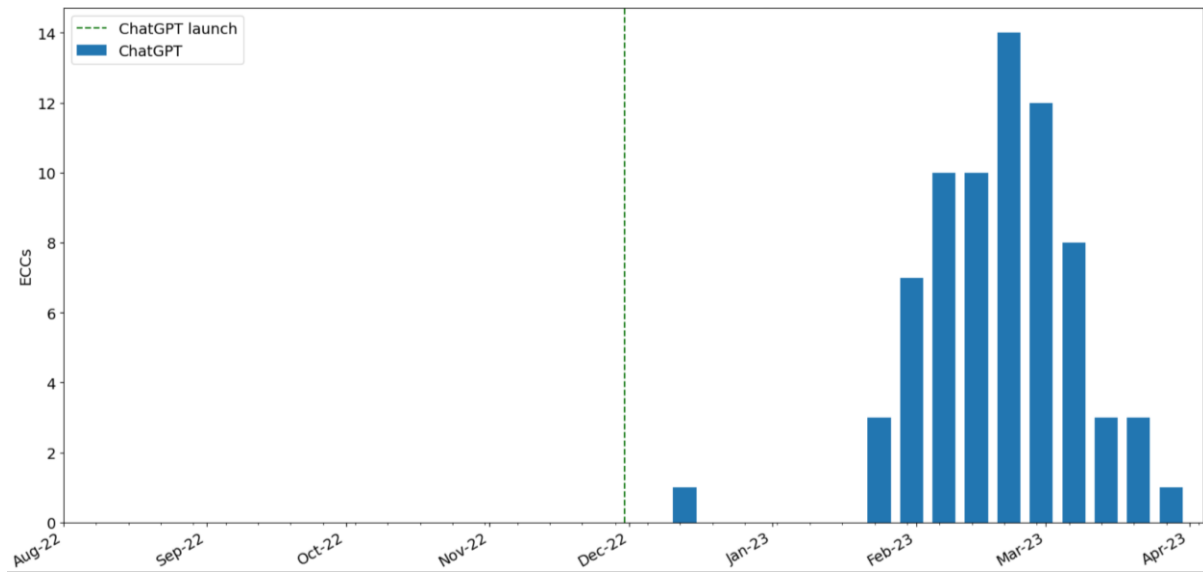
Dep. Var:	ΔAI_ETF	$\Delta COVERAGE$	$\Delta TRAD_VOL$	$\Delta DAYS$	$\Delta CAPEXINT$
	(1)	(2)	(3)	(4)	(5)
<i>SPEC_FIRM</i>	-0.083 (0.220)	2.460 (0.605)	3.090 (0.319)	-4.336 (0.222)	0.002 (0.991)
<i>LAMBDA</i>	-0.185 (0.103)	-6.087 (0.449)	-6.012 (0.238)	13.479** (0.031)	-0.539* (0.084)
Constant	-1.084*** (0.000)	-1.513*** (0.000)	-1.091*** (0.000)	-1.453*** (0.000)	-1.148*** (0.000)
Firm controls	Y	Y	Y	Y	Y
Industry controls	Y	Y	Y	Y	Y
ECC controls	Y	Y	Y	Y	Y
Model type	Heckman	Heckman	Heckman	Heckman	Heckman
N	451	402	449	409	436

p-values in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Appendix 1: The evolution of mentions of “ChatGPT” in public firms’ ECCs

This figure reports the frequency of ECCs which mention ChatGPT in the window four months before the public launch of ChatGPT (on 30 November 2022) and four months afterwards. The vertical dotted line shows the timing of the public launch of ChatGPT, which is the “shock” used in this paper.



Appendix 2: Sample statistics of initial AI ECCs

This table describes our sample of initial AI ECCs by ECC working week and by industry position. Panel A presents the number of initial AI ECCs by working week in the period of interest. Panel B presents the breakdown of all firms in the sample by their distribution across the AI industry value chain. Inclusion in the AI value chain is based on membership of specific 6-digit NAICS codes. Firms not members of those NAICS-based industries are classified as “Users”. Panel C then presents the number of initial AI ECCs for these potential ‘AI beneficiaries’ by Fama–French 12 industries. In each panel, we provide data for Speculative and existing firms, as well as the proportion of all initial AI ECCs in that week/industry. Panel D reports the split of the sample between firms that were S&P 500 constituents on the day ChatGPT was launched. Finally, Panel E reports the split of firms by company life-stage using proxies developed by Dickinson (2011).

	Speculative	Existing	Total	% Speculative
Panel A: By working week				
2022 w/c 28 Nov	1	5	6	16.7%
05 Dec	1	11	12	8.3%
12 Dec	2	7	9	22.2%
19 Dec	0	1	1	0.0%
26 Dec	0	0	0	n.a.
2023 2 Jan	0	2	2	0.0%
9 Jan	1	1	2	50.0%
16 Jan	0	2	2	0.0%
23 Jan	11	20	31	35.5%
30 Jan	14	36	50	28.0%
06 Feb	33	39	72	45.8%
13 Feb	30	43	73	41.1%
20 Feb	40	54	94	42.6%
27 Feb	41	46	87	47.1%
06 Mar	13	24	37	35.1%
13 Mar	13	11	24	54.2%
20 Mar	2	9	11	18.2%
27 Mar	12	24	36	33.3%
Total	214	335	549	28.1%
Panel B: By AI value chain				
User	165	229	394	41.9%
IT hardware manufacture	5	8	13	38.5%
Computer component manufacture	5	18	23	21.7%
IaaS, data process & hosting	23	53	76	30.3%
Software (middleware, apps)	11	21	32	34.4%
IT services	5	6	11	45.5%
Total	214	335	549	39.0%

Panel C: AI beneficiaries by Fama-French industry (12)				
Consumer non-durable goods	6	7	13	46.2%
Consumer durables goods	5	4	9	55.6%
Manufacturing	13	17	30	43.3%
Oil, gas, and coal extraction and products	1	8	9	11.1%
Chemicals and allied products	1	2	3	33.3%
Business equipment and software	34	53	87	39.1%
Telecom and TV	4	5	9	44.4%
Utilities	0	0	0	n.a.
Wholesale, retail and some services	16	20	36	44.4%
Healthcare, medic equip, and drugs	20	31	51	39.2%
Finance	14	12	26	53.8%
Other	25	34	59	42.4%
Total	139	193	332	41.9%

Panel D: Constituent of the S&P 500				
Constituent	40	63	103	38.8%

Panel E: Company life-stage				
Introduction	25	69	94	26.6%
Growth	41	45	86	47.7%
Mature	85	140	225	37.8%
Shake-Out	37	49	86	43.0%
Decline	13	14	27	48.1%
Total	201	317	518	38.8%

Appendix 3: AI disclosure relevance

This table shows the relevance of AI disclosures to changes in analysts' target prices. It presents the results of regressing the indicator variable for AI disclosure (*AI_ECC*) on the change in analysts' target prices following the initial ECC (ΔTP), using OLS. The p-values are shown in parentheses, where ***, **, and * denote the significance of the p-values at 1%, 5%, and 10% levels.

Dep. Var: ΔTP	
Sample:	Main
<i>AI_ECC</i>	2.513*** (0.000)
<i>MARKET VALUE</i>	-0.003 (0.991)
<i>R&D</i>	1.672 (0.833)
<i>Q4</i>	-1.137 (0.255)
<i>ECC_TONE</i>	-2.893* (0.064)
<i>CAR</i>	1.013 (0.830)
Constant	-2.736 (0.342)
Firm controls	Y
Industry controls	Y
ECC controls	Y
Model	OLS
N	1,814
Adj. R-squared	0.008

p-values in parentheses

* p < 0.10, ** p < 0.05, *** p < 0.01

Appendix 4: Linguistic characteristics

This table describes the differences in several linguistic characteristics of AI disclosures by Existing and Speculative firms. Panel A shows several ‘word count’-based measures. Panel B consists of several linguistic ‘quality’ measures from the literature. Definitions of each of these measures are provided in Appendix E. Column (1) provides the sign of the expected difference between Speculative and Existing firms, the result of which is shown in column (4). (2) and (3) show the mean value for each firm type. Column (4) displays the difference between these two means, and column (5) shows the p-value of this difference based on a t-test. The significance of the p-values is shown, where ***, **, and * denote the significance of the p-values at 1%, 5%, and 10% levels.

	Expected sign	Existing Mean	Speculative Mean	Difference Value p-value	
Panel A: Quantitative measures					
<i>AI_WORDS_%</i>	-	0.023	0.014	-0.009	0.010 ***
<i>AI_SENTENCES_%</i>	-	0.024	0.015	-0.009	0.005 ***
<i>AI_KEYTERMS_%</i>	-	0.046	0.045	-0.001	0.803
<i>AI_KEYTERMS_RANGE</i>	-	2.198	1.922	-0.276	0.194
Panel B: Qualitative measures					
<i>SENTIMENT_LM</i>	+	0.326	0.324	-0.002	0.937
<i>SENTIMENT_GPT</i>	+	0.422	0.414	-0.007	0.817
<i>WEASEL</i>	+	0.012	0.014	0.001	0.456
<i>VAGUE</i>	+	0.462	0.606	0.144	0.000 ***
<i>SPECIFIC</i>	-	0.712	0.422	-0.290	0.000 ***
<i>COMPLEX</i>	-	0.084	0.053	-0.031	0.015 **
<i>OBFU</i>	+	0.285	0.294	0.009	0.076 *

Appendix 5: Firm characteristics

This table presents the characteristics of Speculative and Existing firms. Panel A includes general, financial and operational characteristics of firms. Panel B contains measures of firm performance. In some cases, where an expected sign isn't shown, these are included for completeness, rather than because we expect a specific result. Panel C contains factors that intuitively should be associated with 'rule-breaking' firms, or 'true negative' measures of meaningful engagement with AI (and, therefore, potentially related to speculative disclosure). Column (1) provides the sign of the expected difference between Speculative and Existing firms, the result of which is shown in column (4). (2) and (3) show the mean value for each firm type. Column (4) displays the difference between these two means, and column (5) shows the p-value of this difference based on a t-test. The significance of the p-values is shown, where ***, **, and * denote the significance of the p-values at 1%, 5%, and 10% levels.

	Expected sign	Existing firms	Speculative firms	Difference	p-value
	(1)	(2)	(3)	(4)	(5)
Panel A: General					
<i>TOTAL_ASSETS</i>	-	7.542	6.649	-0.893	0.000 ***
<i>REVENUE</i>	-	6.890	5.728	-1.162	0.000 ***
<i>BM</i>	+	0.465	0.621	0.156	0.041 **
<i>DEBT_EQUITY</i>	+	2.186	1.634	-0.551	0.506
<i>AGE</i>	-	20.380	18.493	-1.886	0.457
<i>R&D</i>	+	0.077	0.100	0.023	0.081 *
<i>INTANGIBLES</i>	+	0.342	0.357	0.015	0.532
<i>CR(4)</i>	-	0.537	0.522	-0.015	0.546
Panel B: Performance					
<i>ROA</i>	+	-0.081	-0.163	-0.082	0.027 **
<i>LOSS</i>	+	0.418	0.556	0.137	0.016 **
<i>RETURN</i>	+	-0.310	-0.436	-0.126	0.013 **
Panel C: "Poor management" attributes					
<i>COMMENT</i>	+	1.610	2.111	0.501	0.030 **
<i>LITIGATION</i>	+	0.044	0.022	-0.021	0.346
<i>RESTATE</i>	+	0.366	0.444	0.078	0.348
<i>D_ACCRUALS</i>	+	-0.000	-0.000	-0.000	0.933
<i>CORP_GOV</i>	-	6.692	6.365	-0.327	0.051 *

Appendix 6: Firm's motivational context and potential motives

This table describes the differences in several firms' motivational context and motives. Panel A includes measures of three different contextual factors that may impact a firm's decision about its disclosure concerning AI. These are: i) the level of uncertainty there is in the market and amongst analysts about the prospects for the firm, ii) the level of monitoring of the firm, and iii) the managerial time horizon and manager's potential incentives to manipulate the firm's share price in the short term. Panel B includes measures relating to two specific motives for speculative disclosure: financing and image management. Column (1) provides the sign of the expected difference between Speculative and Existing firms, the result of which is shown in column (4). (2) and (3) show the mean value for each firm type. Column (4) displays the difference between these two means, and column (5) shows the p-value of this difference based on a t-test. The significance of the p-values is shown, where ***, **, and * denote the significance of the p-values at 1%, 5%, and 10% levels.

	Expected sign	Existing Firms	Speculative Firms	Difference	p-value
	(1)	(2)	(3)	(4)	(5)
Panel A: Motivational context					
Uncertainty					
<i>BETA</i>	+	0.33	0.38	0.05	0.392
<i>BID_ASK</i>	+	0.00	0.00	0.00	0.199
<i>TP_IQR</i>	+	0.56	0.60	0.04	0.877
Monitoring					
<i>COVERAGE</i>	-	10.26	7.68	-2.58	0.012 **
<i>INSTITUTIONAL</i>	-	0.66	0.56	-0.10	0.013 **
<i>MEDIA</i>	-	4.02	3.87	-0.15	0.202
<i>AUDITOR</i>	-	0.68	0.57	-0.11	0.047 **
<i>INT_CONTROL</i>	-	0.23	0.26	0.03	0.814
CEO incentives					
<i>EQUITY</i>	+	-0.01	0.11	0.12	0.326
<i>CEO_SBC</i>	+	0.61	0.44	-0.17	0.003 ***
<i>CEO_OPTIONS</i>	+	1.56	8.01	6.45	0.000 ***
Panel B: Motives					
Financing					
<i>EQUITY_RAISE</i>	+	0.04	0.04	-0.00	0.865
<i>FIN_CF</i>	+	0.01	0.03	0.02	0.426
<i>FIN_CONSTR</i>	+	-0.67	-0.20	0.48	0.005 ***
<i>GOING_CON</i>	+	0.03	0.04	0.01	0.623
Image management					
<i>SUE_SCORE</i>	-	1.67	0.84	-0.83	0.264
<i>JUST_MISS</i>	+	0.12	0.15	0.03	0.444

Appendix 7: Results of Propensity Score Matching (PSM)

This table compares the means of firm characteristics for the most recent fiscal year-end, immediately preceding the initial AI ECC, between the Speculative and Existing firms. Panel A compares Speculative and Existing treated firms to illustrate the differences in the means of key firm characteristics of firms in our main sample. Panel B shows the differences between the treated and control groups for both Speculative and Existing firms. This is designed to illustrate the veracity of the matching. Firms disclosing to the market about their AI activities in the initial ECC are considered to be part of the ‘treatment group’. Control group firms that did not mention AI in their initial ECC were matched to one of the treated firms using PSM, with the nearest (single) neighbour method, without replacement. ***, **, and * denote the significance of the p-values (at 1%, 5%, and 10% levels) of the respective differences [columns (5) in Panel A), and columns (4) and (8) in Panel B].

Panel A: Comparison of Speculative and Existing Treated Firms					
	Speculative firms		Existing firms		Difference
	N	Mean	N	Mean	
	(1)	(2)	(3)	(4)	(5)
ΔTP	56	0.01	242	-0.03	0.04
General					
$MARKET_VALUE$	56	7.15	242	8.00	-0.85 ***
$REVENUE$	56	6.12	242	7.07	-0.96 ***
BM	56	0.47	242	0.41	0.06
$DEBT_EQUITY$	56	1.60	241	2.18	-0.58
AGE	56	16.88	241	21.08	-4.20
$R\&D$	56	0.10	242	0.09	0.01
$INTANGIBLES$	54	0.33	220	0.33	0.00
Performance					
ROA	56	-0.12	242	-0.08	-0.05
$LOSS$	56	0.61	242	0.44	0.17 **
$RETURN$	56	-0.43	237	-0.32	-0.11 **
Internal and external monitoring					
IO	56	0.61	242	0.68	-0.07 *
$MEDIA$	54	4.00	233	4.17	-0.16
Demand for capital					
$EQUITY_RAISE$	55	0.02	234	0.04	-0.01
FIN_CF	56	0.02	242	0.01	0.01
FIN_CONSTR	56	-0.44	241	-0.83	0.39 **

Panel B: Veracity of PSM matching								
	Speculative firms				Existing firms			
	N	Treatment	Control	Difference	N	Treatment	Control	Difference
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
ΔTP	56	0.01	-5.06	5.07 **	242	-0.03	-2.27	2.23 *
General								
$MARKET_VALUE$	56	7.15	7.79	-0.64 *	242	8.00	7.58	0.42 **
$REVENUE$	56	6.12	6.99	-0.87 **	242	7.07	6.97	0.10
BM	56	0.47	0.47	0.00	242	0.41	0.52	-0.11 **
$DEBT_EQUITY$	56	1.60	1.89	-0.29	241	2.18	1.78	0.40
AGE	56	16.88	26.73	-9.86 ***	241	21.08	26.21	-5.13 **
$R\&D$	56	0.10	0.09	0.01	242	0.09	0.06	0.03 **
$INTANGIBLES$	54	0.33	0.27	0.06 *	220	0.33	0.28	0.06 ***
Performance								
ROA	56	-0.12	-0.03	-0.09 *	242	-0.08	-0.04	-0.04
$LOSS$	56	0.61	0.36	0.25 ***	242	0.44	0.38	0.06
$RETURN$	56	-0.43	-0.18	-0.25 ***	237	-0.32	-0.25	-0.06 *
Internal and external monitoring								
IO	56	0.61	0.74	-0.13 **	242	0.68	0.77	-0.09 ***
$MEDIA$	54	4.00	3.78	0.23	233	4.17	3.82	0.35 ***
Demand for capital								
$EQUITY_RAISE$	55	0.02	0.02	0.00	234	0.04	0.02	0.01
FIN_CF	56	0.02	-0.02	0.04 *	242	0.01	-0.02	0.03 *
FIN_CONSTR	56	-0.44	-3.89	3.45 ***	241	-0.83	-3.72	2.90 ***

Appendix 8: Descriptive statistics (extended)

This table presents descriptive statistics for the variables used in the models in Tables 5, 6 and 7. The additional variables of interest are used as moderating factors in Table 5. Descriptive statistics for Table 4 are provided in Table 2. Definitions for all variables are provided in Appendix E.

	N	Mean	Standard deviation	Minimum	Maximum
<i>Dependent variables</i>					
<i>CAR[-1, +1]</i>	388	0.009	0.117	-0.420	0.305
<i>CAR[-1, +5]</i>	388	-0.004	0.138	-0.645	0.368
<i>ΔAI ETF</i>	388	0.369	0.623	0.000	2.000
<i>ΔCOVERAGE</i>	388	-31.555	29.312	-83.333	100.000
<i>ΔTRADE_VOL</i>	386	2.273	23.612	-41.489	98.481
<i>ΔDAYS</i>	388	-23.188	24.283	-93.000	37.000
<i>ΔCAPXINT</i>	378	0.440	1.399	-0.808	12.152
<i>SPEC_FIRM2</i>	388	0.173	0.378	0.000	1.000
<i>ΔTP2</i>	375	-0.017	0.182	-0.981	0.921
<i>Variables of interest</i>					
<i>GENEXP</i>	387	359.783	105.742	125.810	657.393
<i>FIRMEXP</i>	387	101.036	69.355	1.476	258.939
<i>NUMFIRM</i>	387	20.181	4.797	6.872	47.882
<i>NUMIND</i>	387	4.074	1.724	1.000	11.000
<i>OPTIMISM</i>	354	0.481	0.063	0.217	0.854
<i>Q4</i>	388	0.624	0.485	0.000	1.000
<i>SUE SCORE</i>	376	1.924	5.872	-22.031	23.005

Appendix 9: Alternative definitions of *SPEC_FIRM* and ΔTP

This table presents the results of robustness tests using alternative definitions of the dependent variable (% change in TP) and the independent variable of interest (*SPEC_FIRM*), compared with the main model. All results are based on Heckman selection correction models, with the second-stage (outcome) equations reported. Models (1) to (2) use the main model (model (2) in Table 4) as a baseline. Model (3) in this table is the same as model (2) in Table 4, reproduced here to aid comparability. The robustness tests are as follows. Model (1) applies an alternative definition of *SPEC_FIRM*. The definition uses an industry-level AI exposure measure (from the Business Trends and Outlook Survey) and an alternative measure of speculative disclosure (the specificity of the AI disclosure content, measured by the *SPECIFIC* variable). Model (2) adjusts the baseline model by substituting a different approach to calculating the percentage change in TP ($\Delta TP2$). The alternative first calculates the percentage difference in the scaled target price at the analyst level, then computes the firm-level value as the median of the analyst-level differences. Standard errors are unadjusted. In all cases, p-values are shown in parentheses, where ***, **, and * denote the significance of the p-values at 1%, 5%, and 10% levels.

	Alternate <i>SPEC_FIRM</i>	Alternate ΔTP	Baseline
Dep. Var.:	ΔTP (1)	$\Delta TP2$ (2)	ΔTP (3)
<i>SPEC_FIRM2</i>	-0.036 (0.238)		
<i>SPEC_FIRM</i>		-0.006 (0.832)	0.024 (0.425)
<i>LAMBDA</i>	-0.002 (0.962)	0.026 (0.578)	-0.005 (0.922)
Constant	-1.625*** (0.000)	-1.680*** (0.000)	-1.625*** (0.000)
Firm controls	Y	Y	Y
Industry controls	Y	Y	Y
ECC controls	Y	Y	Y
Model	Heckman	Heckman	Heckman
N	388	375	388

Appendix 10: Placebo test

This table presents the results of a placebo test compared with the main result. Both results are based on Heckman selection correction models. It shows the results of the second-stage (outcome) equations for each case. The percentage change in target prices is the dependent variable in both models. Model (1) is the placebo test, following the exact specification as the Heckman model presented in Table 4 – model (2). In model (1), the assignment of treatment (*SPEC_FIRM*) is randomised across the same sub-sample of firms, maintaining the same ratio of speculative to existing firms as in the main result. Standard errors are unadjusted. In both panels, p-values are shown in parentheses, where ***, **, and * denote the significance of the p-values at 1%, 5%, and 10% levels.

	Placebo (1)	Baseline (2)
Dep. Var: ΔTP		
<i>SPEC_FIRM</i>	0.000 (0.994)	0.024 (0.425)
<i>LAMBDA</i>	-0.005 (0.919)	-0.005 (0.922)
Constant	-1.625*** (0.000)	-1.625*** (0.000)
Firm controls	Y	Y
Industry controls	Y	Y
ECC controls	Y	Y
Model	Heckman	Heckman
N	388	388

Appendix A: AI dictionary

This table provides the keywords and terms (and abbreviations) used in the AI dictionary. The dictionary is constructed in two steps. Firstly, by combining dictionaries of Jia et al. (2025) and Cao et al. (2024), deduplicating between them, and then adding plural terms and abbreviations of the combined set. Secondly, by augmenting this combined dictionary with AI use cases from (OECD 2022) selected by ChatGPT as relevant and likely to be mentioned in an earnings call.

Terms derived from Jia et al. (2025) and Cao et al. (2024)		Terms selected from OECD (2022)
ai	gpt-4	acoustic model
ai/ml	gpt3	asr
alphafold	gpt3.5	cognitive methods
anthropic	gpt4	edge detection
artificial intelligence	hugging face	evolutionary computation
bard	image recognition	evolutionary strategies
big data	information extraction	expert system
business intelligence	jasper	expert systems
chatbot	kernel method	facial recognition
chatgpt	large language model	fuzzy logic
classification algorithm	large language models	gesture detection
classification algorithms	lightricks	handwriting recognition
claude	llm	inference engine
clustering algorithm	machine learning	inference rules
clustering algorithms	machine translation	language model
cohere	makersuite	neural network hardware
computational linguistic	midjourney	npu
computer vision	ml	object detection
controlnet	natural language generation	probabilistic model
copilot	natural language processing	reinforcement learning
dall-e	neural network	speech recognition
data mining	neural networks	tokenization
data science	nlp	voice assistants
data scientist	object recognition	acoustic models
data scientists	openai	cognitive method
deep learning	palm api	inference engines
deepmind	power bi	language models
diffusion model	prompt learning	probabilistic models
diffusion models	recommender system	voice assistant
dimensionality reduction	recommender systems	
gai	rerank	
gan	stability ai	
gen ai	stable diffusion	
genai	supervised learning	
generative adversarial network	support vector machine	
generative adversarial networks	svm	
generative ai	text to image	
generative pre-trained transformer	transformer	
generative pre-trained	transformers	
transformers	unsupervised learning	
google ai	vae	
google deepmind	variational autoencoder	
gpt	variational autoencoders	
gpt-3	whisper	
gpt-3.5		

Appendix B: Definitions of linguistic measures

This table provides the definitions of linguistic measures. Each metric measures a different attribute of firms' AI disclosure in their initial ECCs. Key terms are words, bigrams, and trigrams related to AI. In the analysis, a word, bigram, and trigram have the same weighting.

Linguistic measure	Definition
<i>AI_KEYTERMS_%</i>	The number of AI-related key terms used by management in the initial ECC as a percentage of all of management's words in the ECC. The raw statistic is then transformed into standard z-scores and shifted to ensure that no values remain non-zero. We then log-transform the result before applying max-min scaling.
<i>AI_KEYTERMS_RANGE</i>	The number of unique AI-related key terms used by management in the initial ECC. The raw statistic is then transformed into standard z-scores and shifted to ensure that no values remain non-zero. We then log-transform the result before applying max-min scaling.
<i>AI_SENTENCES_%</i>	The number of sentences which contain at least one AI key term divided by the total number of sentences in management's dialogue in the ECC transcript.
<i>AI_WORDS_%</i>	The number of words in sentences which contain at least one AI key term divided by the total number of words in management's dialogue in the ECC transcript.
<i>COMPLEXITY</i>	The MODFOG score (Kim et al., 2019) of management's AI disclosure. This is a modified FOG score of the disclosure, which assesses the language for its complexity. The modification involves the exclusion of words deemed common in financial disclosure and, therefore, not complex.
<i>OBFU</i>	Compares the MODFOG score for management for the AI disclosure, as above, to the MODFOG score for the language used by the analysts on the same ECC. The <i>OBFU</i> score is the residual resulting from regressing the MODFOG for the analyst on the MODFOG of the management (Bushee et al., 2017).
<i>SENTIMENT_LM</i>	Firm-level measure of the average sentiment of management's AI dialogue, constructed by asking ChatGPT to assign a sentiment score [positive (1), neutral (0) or negative (-1)], based on Loughran and McDonald's (2011) dictionary of positive and negative words.
<i>SENTIMENT_GPT</i>	Firm-level measure of the average sentiment of management's AI dialogue, constructed by asking ChatGPT to assign a sentiment score [positive (1), neutral (0) or negative (-1)], based on its own pre-training (without any prescriptive dictionary).
<i>SPEC_FIRM</i>	An indicator variable which equals 1 if the firm is deemed to be 'speculative', and zero otherwise (in the latter case the firm is considered an 'existing' firm). The measure equals 1 if the firm is in the bottom tercile in terms of AI 'investment' (Babina et al., 2024), and its AI disclosure is adjudged by ChatGPT to be speculative, based on Cheng et al.'s (2019) definition of speculative disclosure, and AI-adapted examples from Cheng et al. (2019).
<i>SPEC_FIRM2</i>	The alternative definition of the main variable of interest. An indicator variable which equals 1 if the firm is deemed to be 'speculative', and zero otherwise. The measure equals 1 if the firm is in the bottom tercile of AI exposure, as measured at the industry level by the US Census Bureau's Business Trends and Outlook Survey (<i>AI_IND</i>), and if the AI disclosure is not considered specific (<i>SPECIFIC</i> = 0).
<i>SPECIFIC</i>	A ChatGPT-assigned indicator of whether management's AI discussion contains specific AI-related initiatives, utilising the same prompt as Jia et al. (2025). The indicator is 1 if firms mention specific initiatives and zero otherwise.
<i>VAGUE</i>	A ChatGPT-assigned vagueness score of firms' AI sentences following the same prompt as Baker et al. (2024).
<i>WEASEL</i>	The number of words in sentences which contain at least one weasel word (Baker et al., 2024) divided by the total number of words in management's dialogue in the ECC transcript.

Appendix C: Examples of AI disclosures

This table includes examples of AI disclosures by firms' management in the sample of initial ECCs we examined. In some cases, the entire disclosure is shown above. In other cases, e.g., Alphabet Inc., these are only extracts from the entire AI disclosure. Panel A (B) provides examples of disclosures by Existing (Speculative) firms.

Company	ECC date	Excerpts from AI disclosure of firms' initial ECCs
Panel A: Existing firms		
Adeia Inc.	22/02/23	"Our core technology blocks such as user interfaces, computer vision, machine learning and networking power a much bigger set of features, applications, services and products, which are deployed across the verticals we target."
AdTheorent Holding Company Inc.	02/03/23	"We also had our sixth consecutive BIG Innovation Award for our privacy-forward machine learning platform, our fifth consecutive win of the AI Breakthrough Award for machine learning, our third consecutive MarTech Breakthrough Award for programmatic marketing innovation and our ninth consecutive year ranking in Crains New York Best Places to Work., Our strong balance sheet and profitability allowed us to continue investing behind products and capabilities that expand our technological advantage over other DSPs.', , 'And we have made a number of enhancements to our visitation models that are driving even greater campaign performance., Our ML models now distinguish density differences in rural versus metropolitan areas to ensure physical visits are most accurately attributed back to ad served impressions.' 'Finally, we introduced sentiment analysis into our platforms targeting and ML modelling logic.', 'We will also launch augmented reality ad units, which enable us to provide interactive ad experiences with virtual information placed in a real-world environment, allowing shoppers to try on sunglasses or hats, for example.', 'Since we started building our award-winning platform in 2012, our focus has been the brain, VML engine, operationalizing complex data science processes, algorithm creation and automation, platform optimizers and more."
Alphabet Inc.	02/02/23	"First, how we unlock the incredible opportunities AI enables for consumers, our partners and for our business; and second, how we focus our investments and make necessary decisions as a company to get there., First, the AI opportunity ahead.', 'AI is the most profound technology we are working on today.', 'Our talented researchers, infrastructure and technology make us extremely well positioned as AI reaches an inflection point.', 'Since then, we have been a leader in developing AI.', 'In fact, our Transformer research project and our field-defining paper in 2017 as well as our pathbreaking work in diffusion models are now the basis of many of the generative AI applications you're starting to see today., Translating these kinds of technical leaps into products that help billions of people is what our company has always strived on.', 'Well pursue this work boldly but with a deep sense of responsibility with our AI principles and the highest standards of information integrity at the core of all our work., We have been preparing for this moment since early last year, and you're going to see a lot from us in the coming few months across 3 big areas of opportunity.', 'First, large models, we published extensively about LaMDA and PaM, the industry's largest, most sophisticated model plus extensive work at DeepMind."

Panel B: Speculative firms

Airbnb Inc.	14/02/23	“And we just take a full-funnel approach to marketing around PR, and we think of our general advertising as really educating people on new products., Now, as far as the changing landscape for technology, I’m actually very excited about the possibilities of AI.', 'And I think that AI is going to really benefit our long tail of data.', 'And I think that AI is going to be a really good opportunity for us.”
Digital Turbine Inc.	08/02/23	“And we can further leverage our on-device position with Ignite to drive AI and machine learning to focus on the right apps to feature on the device versus the customer being overwhelmed by being dumped into a big app store with many millions of apps to choose from.”
BSQUARE Corporation	07/03/23	“As you know, artificial intelligence, machine learning and the models of that are absolutely part of the things that we get involved in with our customers with Square One, some of the larger Edge to Cloud customers., So yes, we do see there’s opportunity for that.', 'I think ChatGPT is getting a lot of buzz right now, and that’s really in AI and putting together words and phrases and sort of answering questions.', 'But absolutely, the use of AI, machine learning and that kind of those kinds of models are things that were involved in, and we’ll continue to look for ways to take advantage of that.”

Appendix D: Definition of variables

This table lists control and dependent variables used in the regressions. A selection of these variables was also used in the Propensity Score Matching. Where applicable, Compustat data item mnemonics and formulas are provided in parentheses

Panel A: Independent/ Control variables	
Variable name	Definition
<i>AGE</i>	The firm's age as of the most recent fiscal year-end.
<i>AIE</i>	Firm-level measure of its 'investment' in (exposure to) AI, calculated as the percentage of employees in the firms with AI skills (Babina et al., 2024).
<i>AI_ECC</i>	An indicator variable which equals 1 (otherwise 0) if the firm mentioned an AI keyword in its initial ECC.
<i>AI ETF</i>	The number of AI-specific exchange-traded funds in which a firm's stock is included.
<i>AI_IND</i>	This is an industry-level (2-digit NAICS) measure of the AI exposure. It is calculated as the percentage of firms in the industry that self-report using AI in the production of goods or services in the previous six months (Question 1 in the AI Supplement). Data source: (US Census Bureau's Business Trends and Outlook Survey – AI Supplementary Tables).
<i>AI_PATENT</i>	The natural logarithm of 1+ the number of AI patents granted to a firm in a three-year window before the launch of ChatGPT. Each firm's three-year window is constructed to incorporate a one-month 'air gap' preceding the earnings reporting cycle prior to the launch of ChatGPT. This is to ensure that news of any AI patents granted to a firm is already incorporated into analysts' target price estimates before we measure the change in target price estimates pre/post initial ECC. The classification of whether a patent relates to AI is probabilistic. We use the measure (variable name: predict93_any_ai) from the US Patent & Trademark Office, with the highest level of probability threshold (93%), as detailed in Pairolero et al (2025).
<i>AI_VC</i>	An indicator variable which equals 1 (otherwise 0) if the firm's 6-digit NAICS code falls within one of the following industries: 334111, 334112, 334118, 334413, 518210, 513210, 541511, 541512, 541611, 541690. These are considered to constitute the AI value chain.
<i>AUDITOR</i>	Indicator variable equal to one if the firm hires a big-four auditor for the most recent fiscal year-end. Data source: Audit Analytics.
<i>BETA</i>	An indicator variable which equals 1 (otherwise 0) if the firm's stock beta calculated over the 30 days prior to the firm's initial ECC, is in the top tercile. The CRSP US Large Cap Index (INDNO: 1001455) is used as the market model.
<i>BM</i>	The firm's book value of total common equity divided by the market value of equity as of the most recent fiscal year-end ($ceq/(csho \times prcc_f)$)
<i>BUS_DAYS</i>	The number of business days in the earnings announcement 'window' since the launch of ChatGPT.
<i>CAPXINT</i>	Capital expenditure intensity. The gross expenditure on tangible assets scaled by total assets for the most recent fiscal year ($capx/at$)
<i>CEO_OPTIONS</i>	The market value of 'in-the-money' exercisable options held by the firm's CEO, scaled by their total compensation in the most recent fiscal year. Data source: Execucomp.
<i>CEO_SBC</i>	The fair value of stock-based compensation granted to the firm's CEO, scaled by their total compensation in the most recent fiscal year. Data source: Execucomp.
<i>COMMENT</i>	The total number of Comment Letter 'threads' initiated by the SEC with the firm in the most recent three years. Data source: Audit Analytics.

<i>CORP_GOV</i>	The firm's Bloomberg Corporate Governance Score for the most recent year.
<i>COVERAGE</i>	The total number of analysts covering the firm during the most recent fiscal year.
<i>CR(4)</i>	Concentration ratio. The proportion of the total sales of firms ranked from first to fourth place in sales within the same industry scaled by the industry's total sales, where industry is defined by its 3-digit NAICS code.
<i>DAYS</i>	The average number of days between the initial ECC and publication of the analysts' next target price forecast
<i>DEBT_EQUITY</i>	The firm's total liabilities divided by its shareholder equity at the most recent year-end (lt/ceq).
<i>D_ACCRUALS</i>	An indicator variable for discretionary accruals (based on Kothari et al. (2005)), which equals 1 (otherwise 0) where the z-score of the modelled discretionary accruals is greater than two.
<i>ECC_TENSE</i>	The 'net tense' of words (measured using the spaCy Natural Language Programming model) used by the firm's managers across the non-AI-related parts of the ECC. Net tense is calculated as the balance of the number of words with a future tense minus the sum of the number of words in the past and present tenses, all divided by the total number of words. Only words where the tense has been determined are included in this calculation.
<i>ECC_TONE</i>	The tone (measured using Loughran and McDonald (2011)) used by the firm's managers across the non-AI-related parts of the ECC.
<i>EQUITY</i>	Proxy for managerial time horizon (Glaeser et al., 2020). Percentage increase in shareholders' equity in the most recent fiscal year ($ceq_t - ceq_{t-1}$)/ ceq_{t-1} .
<i>EQUITY_RAISE</i>	The firm's sale of common and preferred stocks scaled by total assets as of the most recent fiscal year-end (sstk/at).
<i>FEXP</i>	Firm-specific experience of the analyst (Bilinski et al., 2025) measured as the number of weeks since the analyst's first target price estimate was made for the focal firm.
<i>FIN_CONSTR</i>	The firm's financial constraint index (Hadlock and Pierce 2010), calculated as $-0.737 \times \text{Size} + 0.043 \times \text{Size}^2 - 0.04 \times \text{Firm Age}$, as of the most recent fiscal year end.
<i>FIN_CF</i>	The firm's net cash flow from financing activities scaled by total assets as of the most recent fiscal year-end (fincf/at).
<i>GENEXP</i>	General experience of the analyst (Bilinski et al., 2025), measured as the number of weeks since the analyst's first target price estimate was made for any firm
<i>GOING_CON</i>	Indicator variable equal to one if the firm receives a going-concern opinion for the most recent fiscal year-end report. Data source: Audit Analytics
<i>IO</i>	The percentage of a firm's shares held by institutional investors at the end of the most recent calendar quarter prior to the most recent fiscal year-end.
<i>INT_CONTROL</i>	The firm's number of internal control weaknesses as of the most recent fiscal year-end. Data source: Audit Analytics
<i>JUST_MISS</i>	An indicator variable for whether a firm narrowly missed meeting analysts' expectations for EPS in the previous quarter. The variable equals 1 if the <i>SUESCORE</i> (see below) is less than 0 or greater than or equal to 1, and 0 otherwise.
<i>LITIGATION</i>	The total number of lawsuits initiated against the firm in any one of the following categories (class action, accounting, SEC law, financial reporting, shareholder suit, derivative) in the most recent three years, where the court has found in favour of the plaintiff. Data source: Audit Analytics.

<i>LOSS</i>	An indicator equal to 1 if the firm's income before extraordinary items is negative and 0 otherwise as of the most recent fiscal year-end.
<i>MARKET_VALUE</i>	The natural logarithm of 1+ the firm's market value of equity plus total assets minus the book value of total common equity as of the most recent fiscal year-end ($csho \times prcc_f + at - ceq$).
<i>MEDIA</i>	The firm's total number of media items from Credit Text Mining T12M News Count during the last 12 months
<i>NUMFIRM</i>	The number of firms for which the individual analyst issued a target price in the six months leading up to the launch of ChatGPT.
<i>NUMIND</i>	The number of two-digit SIC industries covered by target price estimates issued by analysts.
<i>OPTIMISM</i>	Analyst optimism adapted from the equivalent measure for optimism with respect to earnings forecast by (Hong and Kubik, 2003). It measures firm-level analyst optimism by averaging across covering analysts the frequency with which each analyst's target price exceeds the contemporaneous peer consensus.
<i>Q4</i>	An indicator variable that equals 1 if the ECC relates to the final quarter of the firm's financial year-end and 0 otherwise.
<i>RESTATE</i>	The number of times the firm has had to restate their accounts due to accounting errors or fraud (excluding clerical errors), where the impact of the restatement is adverse for the firm over the most recent five years.
<i>RETURN</i>	The equity returns of the company for the 12 months preceding the release of the initial
<i>REVENUE</i>	The natural logarithm of 1 + the firm's sales as of the most recent fiscal year-end ($revt$).
<i>ROA</i>	The firm's income before extraordinary items divided by lagged total assets as of the most recent fiscal year-end (ib/at).
<i>R&D</i>	R&D intensity. The firm's R&D expenditure divided by total assets as of the most recent fiscal year-end (xrd/at)
<i>SUE_SCORE</i>	Calculated by IBES as the difference between the actual EPS and the mean analyst EPS forecast, scaled by the standard deviation of analysts' EPS forecasts.
<i>TOTAL_ASSETS</i>	The firm's book value of total assets as of the most recent fiscal year-end (at).
<i>TP_IQR</i>	The interquartile range of the Target Prices provided by analysts following the firm's stock.

Panel B: Dependent variables	
Variable name	Definition
<i>ΔCAPEXINT</i>	The percentage change in Capex intensity between the fiscal year prior to the initial ECC and the next fiscal year.
<i>CAR[-1, +1], CAR[-1, +5]</i>	Cumulated Abnormal Return is the sum of abnormal returns over trading days -1 to +1(+5) relative to the initial ECC. Abnormal returns are computed using a market model (based on the CRSP US Large Cap Index (INDNO: 1001455)) estimated over a 180-trading-day estimation window ending five trading days prior to the event, requiring at least 100 valid observations.
<i>ΔCOVERAGE</i>	The change in the number of analysts covering the stock in the three months following the initial ECC date.
<i>ΔDAYS</i>	The change in the number of analyst days required to submit an updated target price (whether the same or revised) between the previous ECC and the initial ECC windows.

ΔAI_ETF	The change in the number of AI-specific exchange-traded funds in which a firm's stock is included between the initial ECC date and the month before their next ECC date.
ΔTP	The percentage change in the firms' mean scaled target price following the initial ECC. The change is calculated for each analyst per firm between the target price submitted immediately prior to the initial ECC and the next target price they submit (whether the same or revised).
$\Delta TP2$	Alternative calculation for the firm-level measure of percentage change in analyst target price following the firm's initial ECC. Calculated as the median of the percentage change in target prices following the initial ECC across all analysts covering the firm
$\Delta TRAD_VOL$	Calculated as the percentage change in the rolling standard deviation of the log of daily share turnover (cshtrd/cshoq) between a pre- and post-event window, where a firm's initial ECC date is considered Day 0. The pre-period window is trading days [-60, -5]. The post-event window is [+2, 30].
